QUALIFICATION STANDARD AND PLAN (QSP)

Marine/Combat System Engineer (MS/NCS Eng)
Naval Engineering Indoctrination (NEI)
NQual (N/A)

MITE ID
Residential Phase – 109286
Afloat Phase – 123457

Regular Force: MS/NCS Eng (00345/00344)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): Commanding Officer HMCS Venture

MARPAC RDIMS # 1340081
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0			Cdr J.J.P. Lebel/CFNES CMDT
1.2			Capt(N) M. Drews/CNPTG
2.0	New QSP Format	27 Jul 23	CO HMCS Venture

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification;

   c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification;

   d. A-PD-055-002/PP-001 NCS Eng Job Based Occupational Specification;

   e. A-PD-055-002/PP-001 MS ENG MOSID 00345 Job Based Occupation Specification for Marine Systems Engineering Officer and

   f. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

3. This QSP publication is effective on receipt and supersedes previous QSP revision 1.2. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened in 2007 at CFNES, CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened in 2007 at CFNES, CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the NEI graduate to perform their duties as a Marine/Combat System Engineer. Possession of this qualification is comprised of knowledge tasks only and does not result in occupational competency.

GRANTING OF QUALIFICATION

2. A trainee will achieve the MITE qualification 109286, Naval Engineering Indoctrination when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. N/A

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. be a Marine System Engineer (00345) or a Naval Combat Systems Engineer (00344); and,

   b. be Naval Warfare Officer II course MITE 119728 qualified.

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 – DESCRIBE THE NAVAL ENGINEERING PROFESSION

1. Performance: Describe the Naval Engineering Profession

2. Conditions:

   a. Given:

      (1) References; and

      (2) Assistance.

   a. Denied:

      (1) Nil.

   b. Environment:

      (1) At sea or alongside in any weather conditions.

3. Standard: IAW Para 4 the Naval Technical Officer shall demonstrate knowledge of the Naval Engineering Profession by describing:

   a. The development of the Naval Engineering Occupations

   b. The history of Naval Engineering;

   c. The career progression of the Naval Engineering Occupation & Roles of Naval Technical NCMs.; and

   d. The role of Naval Engineering Officers.

4. Reference: A4, A5, A6, A7, A8, C01,.C02, D01, A60, C06, C05, C07, A74, A75, B04, C09, C10, C11, A80, A81, A83, D02, D03, C14, C18, C19, A98, C21, C23

5. Task Numbers: K0015, K0020, K0065, K2655, K2675, K2680.

6. Remarks: This course is subject to availability of ship space for the completion of OJPR. The ashore phase may be disrupted due to ship schedules. Candidates also conduct OJPR at sea or alongside on various ships.

PO 002 – DESCRIBE THE OPERATION AND MAINTANCE OF SHIPBOARD SYSTEMS

1. Performance: Describe the Operation and Maintenance of Shipboard Systems.

2. Conditions:

   a. Given:

      (1) References; and

      (2) Assistance.

   b. Denied:

      (1) Nil.

   b. Environment:

      (1) At sea or alongside in any weather conditions.

3. Standard: IAW Para 4 the Naval Technical Officer shall demonstrate knowledge of the Operation and Maintenance of Shipboard systems by describing:

   a. Missions Assigned to Warships;

   b. Factors Affecting Habitability;

   c. Survivability and Damage Limiting Considerations;

   d. Signature Management Requirements;

   e. The Shipboard Maintenance Organization;

   f. Shipboard Engineering Documentation;

   g. Docking and Undocking Considerations; and

   h. Presentation of Technical Information.

4. Reference: A12, A13, A15, A38, A61, A62, A63, A65, A66, B02, B03, B05, A85, A86, A87, A98, C22

5. Task Numbers: DT0010, DT0015, DT0020, DT0035, K0005, K0010, K0030, K0035, K0040, K0045, K0070, K0085, K0090, K0115, K0140, K0145, K0150, K0155, K0160, K0165, K0170, K0175, K0180, K0185 K0190, K0195, K0200, K0205, K0210, K0215, K0220, K0225, K0240, K0245, K0250, K0255, K0260, K0285, K0290, K0300, K0310, K0315, K0330, K0360, K0365, K0370, K0410, K0415, K0430, K0445, K0450,

K0465, K0475, K0480, K0485, K0490, K0495, K0500, K0510, K0515, K0520, K0525, K0530, K0535, K0540, K0545, K0550, K0550, K0560, K0565, K0580, K0595, K0605, K0615, K0625, K0630, K0635, K0650, K0655, K0660, K0665, K0670, K0675, K0680, K0690, K0695, K0700, K0705, K0710, K0715, K0720, K0725, K0740, K0745, K0750, K0755, K0795, K0805, K0860, K0905, K0955, K0960, K1005, K1020, K1030, K1035, K1040, K1045, K1055, K1065, K1070, K1080, K1090, K1095, K1100, K1105, K1110, K1115, K1120, K1125, K1130, K1140, K1145, K1150, K1155, K1160, K1165, K1170, K1175, K1180, K1190, K1210, K1215, K2070, K2090, K2505, K2510, K2530, K2535, K2540, K2550, K2555, K2560, K2685, K2690, K2691.

6. Remarks: This course is subject to availability of ship space for the completion of OJPR. The ashore phase may be disrupted due to ship schedules. Candidates also conduct OJPR at sea or alongside on various ships.

PO 003 – DESCIBE THE OPERATION OF MARINE SYSTEM ENGINEERING EQUIPMENT

1. Performance: Describe the Operation of Marine Systems Engineering equipment.

2. Conditions:

   a. Given:

      (1) References; and

      (2) Assistance.

   c. Denied:

      (1) Nil.

   b. Environment:

      (1) At sea or alongside in any weather conditions.

3. Standard: IAW Para 4 the Naval Technical Officer shall demonstrate technological knowledge of the Operation of Marine Systems Engineering equipment by describing:

   a. Hull Structures;

   b. Main Propulsion;

   c. Ancillary System;

   d. Machinery Control Systems (MCS);

   e. Electrical Power Generation and Distribution Systems (EPG&D); and

   f. Marine Auxiliary Systems.

4. Reference: A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20.

5. Task Numbers: K0075, K0080, K0095, K0100, K0110, K0135, K0140, K0145, K0150, K0155, K0160, K0160, K0190, K0195, K0200, K0210, K0215, K0220, K0230, K0235, K0240, K0245, K0250, K0255, K0260, K0265, K0270, K0280, K0285, K0290, K0295, K0320, K0325, K0340, K0345, K0350, K0355, K0360, K0380, K0390, K0395, K0400, K0405, K0425, K0435, K0460, K0470, K0475, K0480, K0485, K0500, K0515, K0810, K0815, K0820, K0825, K0830, K0840, K0845, K0855, K0860, K0865, K0870, K0880, K0885, K0890, K0895, K0900, K0905, K0915, K0920, K0925, K0930, K0935,

K0940, K0945, K0950, K0965, K0970, K0975, K1015, K1050, K1070, K1075, K1120, K1135, K2005, K2010, K2015, K2020, K2025, K2030, K2035, K2040, K2045, K2050, K2055, K2060, K2065, K2080, K2085, K2500, K2620.

6. Remarks: This course is subject to availability of ship space for the completion of OJPR. The ashore phase may be disrupted due to ship schedules. Candidates also conduct OJPR at sea or alongside on various ships.

PO 004 – DESCRIBE THE OPERATION OF NAVAL COMBAT SYSTEM ENGINEERING EQUIPMENT

1. Performance: Describe the operation of Naval Combat Systems Engineering equipment.

2. Conditions:

a. Given:

(1) References; and

(2) Assistance.

d. Denied:

(1) Nil.

b. Environment:

(1) At sea or alongside in any weather conditions.

3. Standard: IAW Para 4 the Naval Technical Officer shall demonstrate technological knowledge of the Operation of Naval Combat Systems Engineering equipment by describing:

a. Warfare System requirements;

b. Command and Control System (CCS);

c. Naval Weapons Systems;

d. Radar Systems;

e. Sonar Systems;

f. Navigation Systems;

g. Naval Communication Systems; and

h. Electronic Warfare (EW) Systems.

4. Reference: A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22

5. Task Numbers: K0025, K0095, K0110, K0115, K0205, K0235, K0460, K0560, K0790, K0795, K0800, K0805, K0810, K0815, K0825, K0835, K0840, K0845, K0850, K0855, K0865, K0870, K0875, K0880, K0885, K0890, K0895, K0920, K0925, K0930,

K0935, K0940, K0945, K0950, K0980, K0985, K0990, K0995, K1005, K1010, K1025, K1030, K1035, K1040, K1060, K1065, K1070, K1075, K1080, K1085, K1090.

6. Remarks: This course is subject to availability of ship space for the completion of OJPR. The ashore phase may be disrupted due to ship schedules. Candidates also conduct OJPR at sea or alongside on various ship.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Ashore Phase;

   b. Afloat Phase;

   c. Pre-Board Phase; and

   d. Board Phase.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 42 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
SLt/Lt(N)	00344	AIRX	1	All
SLt/Lt(N	00345	AIRY	1	All
LCdr	00344/00345	AISF/AISE	1	PC 002
Lt(N)	00344	AIRX	1	PC 002
Lt(N)	00345	AIRY	1	PC 002

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitation is identified below:

a. Intent of this course is to provide students the sea experience necessary to be successful at the next stage of NTO occupational training. The course includes 25 days aboard an HMC Ship with 15 days of sea time IOT foster appreciation of the operation and maintenance of HMC Ships.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 16;

b. Maximum Load – 30; and

c. Minimum – 6.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. NTO BTL Manager

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-51-Course Reporting and Certificates. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, and leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student’s strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – DESCRIBE THE NAVAL ENGINEERING PROFESSION

EO 001.01 ENABLING OBJECTIVE

1. Performance: Describe the Development of the Naval Engineering Occupations.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW references listed in Para 4, the trainee shall demonstrate knowledge of the development of the Naval Engineering Occupations by describing:

   a. The RCN Engineering Branch as an extension of the RN;

   b. The Electrical Branch;

   c. The Ordinance Branch;

   d. The Constructor Branch; and

   e. Describing the events leading to the creation of the Naval Engineering Officer Occupations.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the RCN Engineering Branch as an extension of the RN**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	**C01**
2	**Describe the Electrical Branch**			
3	**Describe the Ordinance Branch**			
4	**Describe the Constructor Branch**			
5	**Describe the events leading to the creation of the Naval Engineering Officer Occupations.**			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 001 – DESCRIBE THE NAVAL ENGINEERING PROFESSION

EO 001.02 DESCRIBE THE HISTORY OF NAVAL ENGINEERING

1. Performance: Describe the History of Naval Engineering.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the effect of technical advancement and significant events in the history of Naval Engineering by describing:

   a. The historical impact of technical developments had on Naval Engineering;

   b. The historical impact of domestic events had on Naval Engineering; and

   c. The historical impact of significant international events had on Naval Engineering;

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the historical impact that each of the following technical developments had on Naval Engineering**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	**C02, D01, C05, C06, C07, C09, C10, C11, D02, D03, C14, C18, C19, C21, C23**
1.1	Wind powered sailing ships to steam driven ships			
1.2	Steam driven to diesel driven ships Barrel/Round Physical properties;			

Teaching Points		Instructional Strategies	Time (min)	References
1.3	Diesel to gas turbine			
1.4	Computer control			
1.5	Integrated systems			
1.6	Radar			
1.7	Sonar			
1.8	Advanced Weaponry			
2	**Describe the historical impact that each of the following significant domestic events had on Naval Engineering**			
2.1	HMCS BONAVENTURE's confined space incident			
2.2	HMCS SAGUENAY's gun explosion			
2.3	HMCS KOOTENAY's gear box explosion			
2.4	HMCS KOOTENAY's bow replacement			
2.5	HMCS CHICOUTIMI's fire			
2.6	HMCS PROTECTEUR's fire			
3	**Describe the historical impact that each of the following significant international events had on Naval Engineering**			
3.1	The sinking of HMS SHEFFIELD			
3.2	The sinking of the R.M.S Titanic			SOLAS
3.3	USS STARK			
3.4	USS VINCENNES			
3.5	USS THRESHER			SUBSAFE
3.6	LIBERTY SHIPS (WWII)			STRUCTURAL FAILURES
3.7	HNoMS Helge Ingstad			Chain of Accident Causation, Training Failures

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 4 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia Classroom.

7. Remarks: Nil

PO 001 – DESCRIBE THE NAVAL ENGINEERING PROFESSION

EO 001.03 DESCRIBE THE CAREER PROGRESSION OF THE NAVAL ENGINEERING OCCUPATIONS

1. Performance: Describe the Development of the Naval Engineering Occupations.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the the Career Progression of the Naval Engineering Occupations by describing:

   a. NTO training progression;

   b. The Canadian Forces Junior Officer Development Program;

   c. NTO professional development;

   d. NTO career management roles and responsibilities; and

   e. The NCM trades within the MSE & CSE department.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**Describe training progression to include the four developmental periods**	**Method:** Interactive Lecture/ Multimedia /	**150**	**A5, A6, A7, A8, A60, A74, A75, B04, A98**
2	**Describe the Canadian Armed Forces Junior Officer Development Program (CAFJOD)**			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Describe professional development courses (Non-occupation specific job related) such as Instructor courses, LCC, Mental Health awareness, etc**	Classroom discussion **Media:** PowerPoint **Environment:** Classroom		
4	**Describe career management roles and responsibilities**			
5	**Describe the NCM trades within the MSE & CSE departments**			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout;

   b. Multimedia Classroom; and

   c. Video documentary “Bringing PROTECTEUR Home”

7. Remarks: Nil.

PO 001 – DESCRIBE THE NAVAL ENGINEERING PROFESSION

EO 001.04 DESCRIBE THE ROLES OF NAVAL ENGINEERING OCCUPATIONS

1. Performance: Describe the Roles of Naval Engineering Occupations.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the roles of Naval Engineering Occupations by describing:

   a. The roles and responsibilities of Naval Engineers; and

   b. The relationship between command, commissioned members and non-commissioned members.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the role of the Naval Engineer with respect to:**	**Method:** Interactive lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**150**	**A4, A7, A80, A81, A83, A98**
1.1	The responsibilities of a Naval Engineer on board ship			

Teaching Points		Instructional Strategies	Time (min)	References
1.2	The four principal dimensions of Naval Engineering; 1. Operations and maintenance; 2. Design and acquisition; 3. Training; and 4. Infrastructure sustainment.			
2	**Describe the functional relationships between:**			
2.1	Command			
2.2	Other Officer Occupations			
2.3	NCM's			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: EO time is broken down into 1 lecture period plus 2 periods Fireside chat with the Cmdt.

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.01 DESCRIBE THE MISSIONS ASSIGNED TO WARSHIPS

1. Performance: Describe the Missions Assigned to Warships.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the missions assigned to warships by describing:

   a. The concept of a warship;

   b. The typical mission assigned to warship; and

   c. The system-based structure established to support the warship.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define integrated weapon system and show how a RCN Ship is a representative example**	**Method:** Interactive Lecture/ Visit of an HMCS Platform and FMFCB/CS **Media:** PowerPoint	**400**	**A62, B02, B03, B05, C22**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe typical ship missions, including:**			
2.1	Flag Ship			
2.2	Area defense			
2.3	Surface and air warfare			
2.4	Anti-submarine warfare			
2.5	Mine counter-measures			
2.6	Force sustainment			
3	**Describe the system-based structure established to support ship missions (using the float-move-fight maxim) listing principal roles, including:**			
3.1	Hull Systems (float)			
3.2	Marine Systems (move)			
3.3	Combat Systems (fight)			
3.4	1. Warfare suite; 2. Communications suite; and 3. Navigation suite.			

5. Total EO Time: Total: 8 x 50 minute periods as follows:

   a. 8 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks:

   a. The systems approach to engineering emphasizes a top-down approach to problem solving characterized by:

      (1) Goal definition;

      (2) Identification of potential solutions (design studies);

      (3) Feasibility analysis;

      (4) Accommodation of constraints (both system constraints, i.e. Physical/practical/task/role oriented, and imposed, i.e. Budgetary/ political); and

      (5) Validation.

   b. The examples listed are intended to illustrate that an integrated weapon system is not defined by the number of personnel that operate it, rather it is defined by the function it performs (namely, the neutralization of hostile targets); and

   c. Given the level of complexity associated with integrated weapon systems, systemic interdependencies cannot be overemphasized and must be continually considered and understood.

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.02 DESCRIBE FACTORS AFFECTING HABITABILITY

1. Performance: Describe Factors Affecting Habitability.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the factors affecting habitability by describing:

   a. The definition of habitability as it applies to ships;

   b. The environmental factors affecting habitability; and

   c. The effect of shipboard facilities on quality of life.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define Shipboard habitability**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**50**	**A65**
2	**Describe environmental factors including:**			

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Heating, ventilation and air conditioning (HVAC)			
2.2	Noise			
2.3	Vibration			
2.4	Lighting			
2.5	Access – passageways deckheads and paneling			
2.6	Paint schemes			
2.7	Deck coverings			
2.8	Ship's husbandry			
3	**Describe the effect of shipboard facilities on quality of life, including:**			
3.1	Accommodations			
3.2	Lounge and recreation			
3.3	Physical conditioning			
3.4	Offices			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: To get students actively involved in the design of a hypothetical warship, an interactive lecture is recommended to promote participation and productivity.

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.03 DESCRIBE THE SURVIVABILITY AND DAMAGE LIMITING CONSIDERATIONS

1. Performance: Describe the Survivability and Damage Limiting Considerations.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the survivability and damage limiting considerations by describing:

   a. The definition of survivability;

   b. Survivability design considerations based on a warship's intended mission(s); and

   c. Technology used for monitoring and controlling damage.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define survivability concepts to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**50**	**A13,A15**

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Susceptibility to detection and tracking			
1.2	Vulnerability to damage			
2	**Describe the survivability design considerations based on a warship's intended mission(s)**			
3	**Describe damage containment/ minimization to include:**			
3.1	Redundancy			
3.2	Arrangement (e.g. Modularization, utilization, separation, location, etc…)			
3.3	Damage Control (DC) systems and equipment			
3.4	Describe the citadel and ventilation control			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: Ni

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.04 DESCRIBE THE SIGNATURE MANAGEMENT REQUIREMENTS

1. Performance: Describe the Signature Management requirements.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of survivability and damage limiting considerations by describing:

   a. The Ship's Signature;

   b. The components of ship's signature;

   c. Shipboard Signature reduction Measures; and

   d. The Signature Management Organization.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define Ship's Signature**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**100**	**A63,A87**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	Emission sources			
1.2	Quantitatively measured and detected			
2	**Describe the components of a Ship's Signature**			
2.1	Acoustic			
2.2	Electromagnetic – Infrared/ Electro-optic			
2.3	Electromagnetic – Radio Frequency (RF)			
2.4	Radar Cross Section			
2.5	Magnetic			
3	**Describe Reduction Measures**			
3.1	Quiet States			
3.2	EMCON			
3.3	RAM pads			
3.4	Degaussing			
4	**Describe the Signature Management Organization**			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.05 DESCRIBE THE SHIPBOARD MAINTENANCE ORGANIZATION

1. Performance: Describe the Shipboard Maintenance organization.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the shipboard maintenance organization by describing:

   a. Roles of the Executive Officer (XO) & Planned Maintenance Coordinator (PMC);

   b. Organization of the Engineering Departments; and

   c. Defense Resource Management Information System (DRMIS).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the role of the Executive Officer (XO) – Planned Maintenance Coordinator (PMC).**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**50**	**A12**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe the organization of the engineering departments to include:**			
2.1	Head of Department (HOD)			
2.2	Chief of Department (CHOD)			
2.3	Departmental Maintenance Coordinators (DMCs)			
2.4	Senior Maintainers			
2.5	Technicians			
3	**Define DRMIS**			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.06 DESCRIBE SHIPBOARD ENGINEERING DOCUMENTATION

1. Performance: Describe Shipboard Engineering Documentation.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of shipboard engineering documentation by describing:

   a. RCN Engineering policies;

   b. Shipboard documentation; and

   c. Shipboard equipment documentation

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe naval documentation and policies to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**50**	**A12, A13, A61, A85, A86, A98.**
1.1	Naval Orders (NAVORDs)			
1.2	MARLANT/MARPACORDS			

Teaching Points		Instructional Strategies	Time (min)	References
1.3	SSOs			
1.4	CFTOs 1. NEM 2. NaMMS 3. National Publications Database			
2	**Describe ship specific policies**			
2.1	Captain's Standing Orders			
2.2	Captain's Ship Book			
2.3	MSEOTI's			
2.4	CSEOTI's			
3	**Describe general documentation**			
3.1	Handbooks			
3.2	Manuals			
3.3	Drawings			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.07 DESCRIBE DOCKING AND UNDOCKING CONSIDERATIONS

1. Performance. Describe docking and undocking considerations.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of docking and undocking considerations by:

   a. Identifying occasions for docking a ship;

   b. Identifying types of docks;

   c. Identifying personnel required for docking; and

   d. Describing the procedures for docking and undocking a ship.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Identify occasions for docking a ship**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**50**	**A13, A66**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe the different types of docks**			
3	**List the personnel required for docking**			
3.1	Dock Master			
3.2	Docking Officer			
3.3	Naval Architect			
3.4	MSEO			
4	**Describe the basic procedures for docking / undocking**			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks:

   a. Lecture to be provided by Fleet Maintenance Facility (FMF) Cape Scott Docking Officer when available.

EO 002.08 PRESENT TECHNICAL INFORMATION

1. Performance. Present Technical Information.

2. Conditions:

   a. Given:

      (1) PP&S (pens, pencils, stationery).

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom; and

      (2) Boardroom.

3. Standard: IAW the references listed in Para 4, the trainee shall present technical information to senior NTOs by:

   a. Promoting the standard of dress and deportment;

   b. Presenting in a professional manner by;

      (1) Using accepted terminology;

      (2) Interacting with board members;

      (3) Maintaining eye contact;

      (4) Maintaining pace of delivery;

      (5) Maintaining volume and clarity of voice;

      (6) Articulating with body language;

      (7) Projecting with confidence and conviction; and

      (8) Maintaining organization;

      (9) Providing valid and logical answers to questions;

(10) Using a board or drawing to assist in presentation; and

(11) Maintaining clarity of drawings.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Dress and deportment**	**Method:** Interactive Lecture / Multimedia / Small group activities **Media:** Video **Environment**: Classroom	**50**	**A9**
2	**Professionalism**			
3	**Use of proper terminology**			
4	**Interaction with board members**			
5	**Eye contact**			
6	**Pace and duration**			
7	**Volume and clarity of voice**			
8	**Body language and movement**			
9	**Confidence and conviction in answers**			

5. Total EO Time: Total: 36 x 50 minute periods as follows:

a. 1 x 50 min period; and

b. 35 x 50 min periods – EC

6. Training and Learning Aids:

a. Student handout.

b. Multimedia Classroom;

c. Boardroom; and

d. Video.

7. Remarks:

a. 30 periods of EC 001 accounts for a week-long period of pre-board preparation.

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.09 DESCRIBE MSE DEPARTMENT WATCHES

1. Performance. Describe MSE Department Watches.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the MSE department watches.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**With the Marine Systems Engineering Officer (MSEO or A/MSEO), observe and participate in the completion of one (1) full set of MSEO daily rounds.**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	**A12, A13, A15, A38, A61, A62, A63, A65, A66, B02, B03, B05, A85, A86, A87, A98, C22**
2	**With the Senior Engineering Watchkeeper (SEWK), stand a minimum of one (1) watch in the Machinery Control Room (MCR) and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
2.1	Flash-up/shutdown			
2.2	Operation:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Purpose and functionality of shipboard marine systems; 2. Direction/supervision of the Engineering Watch; 3. Relationship with the Officer of the Watch (OOW); and 4. Use of Engineering Officer's Technical Instructions (EOTIs)			
2.3	Trouble-shooting/Health Monitoring: 1. Responses to engineering problems/emergencies; and 2. Conduct of Engineering Delaying/Non-Delaying Drills			
2.4	Conduct of Corrective and Preventive Maintenance (CM and PM)			
3	**With the Engineering Watchkeeper (EWK), stand a minimum of two (2) watches in the Machinery Control Room (MCR) and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
3.1	Flash-up/shutdown			
3.2	Operation: 1. Purpose and functionality of shipboard marine systems; 2. Operation of the Machinery Control Console (MCC); and			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Relationship with the Engineering Officer of the Watch (EOOW);			
3.3	Trouble-shooting/Health Monitoring to include responses to engineering problems/emergencies			
3.4	Conduct of Corrective and Preventive Maintenance (CM and PM).			
4	**With Electrical Technician of the Watch (ETOW), stand a minimum of one (1) watch in the Machinery Control Room (MCR) and Machinery Spaces and observe/participate in activities to include**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
4.1	Flash-up/shutdown			
4.2	Operation: 1. Purpose and functionality of shipboard marine systems; 2. Direction/supervision of the Engineering Watch; 3. Relationship with the Officer of the Watch (OOW); and 4. Use of Engineering Officer's Technical Instructions (EOTIs)			
4.3	Trouble-shooting/Health Monitoring: 3. Responses to engineering problems/emergencies; and 4. Conduct of Engineering Delaying/Non-Delaying Drills			

Teaching Points		Instructional Strategies	Time (min)	References
4.4	Conduct of Corrective and Preventive Maintenance (CM and PM)			
5	**With the Engineering Roundspersons, stand a minimum of two (2) watches in the Machinery Control Room (MCR) and Machinery Spaces and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
5.1	Flash-up/shutdown			
5.2	Relationship with the Engineering Officer of the Watch (EOOW);			
5.3	Trouble-shooting/Health Monitoring to include responses to engineering problems/emergencies;			
5.4	Conduct of Corrective and Preventive Maintenance (CM and PM).			
6	**With the steaming watch, stand a minimum of one (1) watch in the MCR to understand the BDCS function of IPMS and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
6.1	Operation: 1. Purpose and functionality of shipboard Damage Control systems; and 2. Operation of Battle Damage Control System (BDCS);			
6.2	Trouble-shooting/Health Monitoring to include responses to Damage Control problems/emergencies.			

Teaching Points		Instructional Strategies	Time (min)	References
7	**With the Damage Control Roundsperson, stand a minimum of one (1) watch conducting Damage Control Rounds and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
7.1	Relationship with the MCR watchkeepers.			
7.2	Trouble-shooting/Health Monitoring to include responses to Damage Control problems/emergencies.			
8	**Presentations of MSE Department personnel, to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
8.1	Marine Systems Engineering Department Overview, including: 1. Overall organization; 2. Roles/responsibilities of key personnel; and 3. Roles/responsibilities of respective sections.			
8.2	Marine Systems Engineering Department Organization, including: 1. Watch Rotation; and 2. On-Watch responsibilities			
9	**Equipment technical/operational narratives: Refer to individual Enabling Objectives of PO 003 for requisite teaching points in**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	

Teaching Points		Instructional Strategies	Time (min)	References
	support of the Marine Systems Engineering equipment.			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.10 DESCRIBE CSE DEPARTMENT WATCHES

1. Performance. Describe CSE Department Watches.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the CSE department watches.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**With the Combat Systems Engineering Officer (CSEO or A/CSEO), observe and participate in the completion of one (1) full set of CSEO daily rounds**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase.**	**A12, A13, A15, A38, A61, A62, A63, A65, A66, B02, B03, B05, A85, A86, A87, A98, C22**
2	**With the Operations Room Technician of the Watch, stand a minimum of one (1) watch and observe/participate in activities to include**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
2.1	Maintenance:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Software/hardware reconfiguration (as applicable) 2. Necessary updates to the equipment fault log.			
2.2	Flash-up/shutdown			
2.3	Monitor/investigate faults (identified through system monitoring at the console and Operations Room personnel reporting)			
3	**Presentations of CSE Department personnel, to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
3.1	Combat Systems Engineering Department Overview, including: 1. Overall organization. 2. Roles, responsibilities and qualifications of key personnel; and 3. Roles/responsibilities of respective sections.			
3.2	Combat Systems Engineering Department Organization, including: 1. Watch Rotation; and 2. On-Watch responsibilities			
4	**Equipment technical/operational narratives: Refer to individual Enabling Objectives of PO 004 for requisite teaching points in support of the Combat Systems Engineering equipment**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.11 DESCRIBE COMBAT DEPARTMENT WATCHES

1. Performance. Describe Combat Department Watches.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the combat department watches.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**For the Naval Communicator (NAVCOMM) Section, stand a minimum of one (1) watch in the Communications Control Room (CCR) and observe/participate in the following:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase.**	
1.1	Flash-up/shutdown: Witness and/or participate in flash-up/shutdown and basic operator functional checks for all fitted communication equipment.			
1.2	Operation:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Purpose and functionality of all shipboard communication systems; 2. Processing of Message Processing System (MPS) messages including inter- ship, ship/shore and fleet broadcast; 3. Purpose and functioning of the Command and Control Monitoring System (CCMS); to include the use of the Communications Plan (COMPLAN)			
1.3	Trouble-shooting/Health Monitoring, to include the use of CCMS functions and/or Test and Measurement System (TMS).			
2	**For the Sonar Operator (Sonar OP) Section, stand a minimum of one (1) watch in the Operations Room and observe/participate in the following:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
2.1	Flash-up/shutdown: Witness and/or participate in flash-up/shutdown and basic operator functional checks for all fitted Sonar equipment.			
2.2	Operation: 1. Purpose and functionality of all shipboard Sonar systems; 2. Acquisition of sound velocity profile; 3. Launching of Sonobuoy(s);			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Role of AESS in ASW; and 5. Role of Threat Motion Analysis (TMA).			
2.3	Trouble-shooting/Health Monitoring			
3	**For the Naval Electronic Sensor (NESOP) Section, Stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
3.1	Flash-up/shutdown: Witness and/or participate in flash-up/shutdown and basic operator functional checks for all fitted Electronic Warfare (EW) and Fire Control (FC) equipment.			
3.2	Operation: 1. Purpose and functionality of all shipboard EW and FC equipment; 1.1. For the Fire Control Equipment: 1.2. Designate a contact from CMS 330 to the Fire Control System; 1.3. Operate the Fire Control System to develop familiarity with the operator display pages and modes of operation; and 1.4. Understand the Fire Control System's role in the guidance of shipboard			

Teaching Points		Instructional Strategies	Time (min)	References
	missiles and guns in the prosecution of targets; 2. For the Electronic Warfare equipment: 2.1. Operate the ESM and ECM Systems to develop familiarity with the operator display pages and modes of operation; 2.2. Understand the Role of ZIPPOs in Anti-Ship Missile Defense (ASMD); 2.3. Understand the Tactical Employment of ESM and ECM systems for a coordinated response in ASMD;			
3.3	Trouble-shooting/Health Monitoring to include Response to system generated faults.			
4	**For the Naval Combat Information Operators (NCIOP) Section, stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
4.1	Flash-up/shutdown: Witness and/or participate in flash-up/shutdown and basic operator functional checks for all fitted Radar and associated systems.			
4.2	Operation:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Purpose and functionality of all shipboard Radar and associated equipment; 2. Understand the role of the radar trackers and track supervisor (including recognized maritime picture compilation and maintenance); 3. Follow the acquisition and track management of several contacts on both the Air Search Radar and the Surface Search Radar; 4. Observe and/or participate in Link 11 operations; 5. Understand the operation and functionality (including the use of the interrogator and transponder); and 6. Develop an appreciation for radar contact symbols			
4.3	Trouble-shooting/Health Monitoring to include response to system generated faults.			
5	**For the Operations Room Officer (ORO), stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
5.1	Operation: 1. Purpose/functionality of Operations Room personnel and equipment; 2. Relationship with the Officer of the Watch (OOW)			

Teaching Points		Instructional Strategies	Time (min)	References
	and Commanding Officer (CO); and 3. Direction/supervision of Operations Room personnel for the safe conduct of shipboard operations;			
6	**For the Operations Room Officer (ORO), stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
6.1	Operation: 1. Understand role/function of Bridge equipment associated with Helicopter operations, weapon systems, navigation, communications and steering/propulsion; 2. Observe the direction/supervision of Bridge on-watch personnel; 3. Relationship with the Engineering Officer of the Watch (EOOW), Operations Room Officer (ORO) and Commanding Officer (CO); 4. Observe shipboard navigation; 5. Understand the hierarchy of shipboard steering positions and the actions			

Teaching Points		Instructional Strategies	Time (min)	References
	to be taken for a Steering Gear Breakdown; 6. Observe the monitoring/operation of the propulsion plant from the Bridge Control Console (BCC); 7. Understand the control hierarchy of the Integrated Platform Management System (IPMS) Human-Machine Interfaces (HMIs), and the circumstances for changing HMI Station In Control (SIC); 8. Understand the normal bridge communications configuration; 9. Understand the role of the OOW in weapon firings and the weapon firing organization; and 10. Understand the role of the OOW and resources required for the conduct of helicopter operations			
7	**For the Operations Room Officer (ORO), stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
7.1	Combat Department Overview, including: 1. Overall organization;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Roles/responsibilities of key personnel; and 3. Roles/responsibilities of respective sections			
7.2	Combat Department At Sea Organization, including: 1. Watch Rotation 2. On-Watch responsibilities.			
8	**For the Operations Room Officer (ORO), stand a minimum of one (1) watch in the Operations Room and observe/participate in activities to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.12 DESCRIBE SHIPBOARD TECHNICAL READINESS

1. Performance. Describe Shipboard Technical Readiness

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the shipboard technical readiness.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Observe and participate in the administration of technical readiness activities and documentation including:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase.**	**A12, A13, A15, A38, A61, A62, A63, A65, A66, B02, B03, B05, A85, A86, A87, A98, C22**
1.1	CSED/MSED Pre-sail checks and reporting			
1.2	CSED/MSED Daily equipment checks			
1.3	CSEO/MSEO Daily Rounds			
1.4	CSEO/MSEO Technical Instructions and Memorandums.			

Teaching Points		Instructional Strategies	Time (min)	References
2	**Oral Presentation: Narrative of Technical Readiness activities and documentation, to include items identified above.**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
3	**Oral presentation of DRMIS and its Significance to Naval Engineering, to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase**	
3.1	Purpose of DRMIS;			
3.2	Usages of DRMIS.			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil

PO 002 – DESCRIBE THE OPERATION AND MAINTENANCE OF SHIPBOARD SYSTEMS

EO 002.13 DESCRIBE NAVAL ENGINEERING LOGISTIC REQUIREMENTS

1. Performance. Describe Naval Engineering Logistic Requirements

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of naval engineering logistic requirements.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of the Supply Department spaces, to include:**	**Method:** OJT **Media:** N/A **Environment:** At Sea or alongside	**Throughout the afloat phase.**	A12, A13, A15, A38, A61, A62, A63, A65, A66, B02, B03, B05, A85, A86, A87, A98, C22
1.1	Observe and participate in the administration and management of shipboard engineering logistic requirements through the conduct of one (1) watch with the Supply Technicians.			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.01 DESCRIBE HULL STRUCTURES

1. Performance. Describe Hull Structures.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of hull structures by describing:

   a. Main components;

   b. Basic stability terms;

   c. Ship loading conditions; and

   d. Liquid level management.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**List the main components that make up the structure of the ship's hull to include**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**A13, A17,C15**
1.1	Plating			

Teaching Points		Instructional Strategies	Time (min)	References
1.2	Keel			
1.3	Longitudinal			
1.4	Beams			
1.5	Decks			
1.6	Bulkheads			
2	**List basic stability terms to include:**			
2.1	Overall Stability			
2.2	Trim			
2.3	List			
2.4	Buoyancy			
2.5	Righting moment			
3	**Describe the ship's loading conditions to include:**			
3.1	Sagging			
3.2	Hogging			
3.3	Racking			
3.4	Slamming			
4	**List the liquid cargo to include:**			
4.1	DFO			
4.2	SWB			
4.3	JP5			
4.4	CFW			
4.5	Why we monitor liquid level			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks:

   a. When available, lectures will be given by a naval architect.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.02 DESCRIBE MAIN PROPULSION SYSTEMS

1. Performance. Describe Main Propulsion Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the main propulsion systems by describing:

   a. Main components;

   b. Prime movers;

   c. Configuration of the transmission system;

   d. Operation of the transmission system; and

   e. Operation of shafting.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Identify the main components of a propulsion system to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**100**	**A18**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	Prime Movers			
1.2	Transmission System			
1.3	Shaftlines			
2	**Describe the prime movers to include**			
2.1	Gas Turbines			
2.2	Diesel Engine			
3	**Describe the operation and configuration of the transmission system to include:**			
3.1	Gearboxes (Main and XConn)			
3.2	Clutches			
3.3	Drive Mode Configurations			
4	**Describe the purpose and operation of shafting, to include:**			
4.1	Shaft line segments			
4.2	Bulkhead / stern seals			
4.3	Thrust bearings			
4.4	Supporting and aligning elements			
4.5	Flexible coupling			
4.6	Propeller (CRPP)			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.03 DESCRIBE ANCILLARY SYSTEMS

1. Performance. Describe Ancillary Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the ancillary systems by describing:

   a. Main propulsion air supply systems;

   b. Main sea water circulating system;

   c. Fuel oil service systems; and

   d. Lubricating oil systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe propulsion air supply systems, to include**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**A16**

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Purpose			
1.2	GT Airflow to include filters/cooling vs combustion, DRES Ball			
1.3	PDE Airflow to include filters, combustion, exhaust			
2	**PDE Airflow to include filters, combustion, exhaust:**			
2.1	Purpose			
2.2	General piping layout, sea chest, pump, cooler, and overboard			
3	**Describe fuel oil service systems**			
3.1	Purpose			
3.2	General layout to include: storage tanks, service tanks, filtration, supply and return lines			
3.3	Emergency arrangements			
3	**Describe lubricating oil systems**			
3.1	Purpose			
3.2	General layout to include: storage tanks, filtration, supply and return lines			
3.3	Emergency arrangements			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.04 DESCRIBE THE MACHINERY CONTROL SYSTEMS

1. Performance. Describe the machinery control systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in para 4, the trainee shall demonstrate technical knowledge of the machinery control systems by describing:

   a. The function and purpose of the MCS;

   b. The design objectives;

   c. The functional capability;

   d. The performance;

   e. The architecture; and

   f. The interfaces with other systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define MCS, to include**	**Method:** Interactive Lecture / Platform visit	**100**	**A41,C08,C20**

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
1.1	Purpose			
1.2	Increased operational capability			
1.3	Manning Reduction			
1.4	Increased control system survivability			
1.5	Increased inter-system capability			
1.6	Embedded operator training capability			
2	**Describe MCS design objectives, to include:**			
2.1	Survivability			
2.2	Reliability			
2.3	Redundancy			
2.4	Modularity			
2.5	Standardization			
2.6	Graceful degradation			
2.7	Flexibility			
3	**Describe MCS capabilities, to include:**			
3.1	System software			
3.2	Operating system			
3.3	Control and monitoring			
3.4	MCS self-diagnostics			
4	**Describe MCS performance, to include:**			
4.1	Closed loop control			
4.2	Operator monitoring			
5	Describe MCS system architecture			
5.1	Remote Terminal Units (RTUs)			
5.2	Engine Control Modules (ECMs)			
5.3	Equipment Health Monitoring (EHM)			
5.4	Human/Machinery Interfaces (HMIs)			
5.5	Configuration Layout			
6	**Describe MCS interfaces with other systems**			
6.1	Damage Control Console (DCC) / IPMS Interfaces – BDCS(DCS)			
6.2	Propulsion Diesel Engine (PDE) and Local Control Cabinets/Panels (LCC/LCPs)			
6.3	Switchboards			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.05 DESCRIBE ELECTRICAL POWER GENERATION AND DISTRIBUTION (EPG&D) SYSTEMS

1. Performance. Describe Electrical Power Generation and Distribution (EPG&D) Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge ofthe Electrical Power Generation and Distribution (EPG&D) Systems by describing:

   a. The configuration of shipboard EPG&D systems;

   b. The power supplied by EPG&D systems;

   c. The categories of power;

   d. The casualty power distribution; and

   e. Power distribution safety concerns.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the configuration major electrical equipment, to include the location of:**	**Method:** Interactive Lecture	**150**	**A15,A26,A27,A39,B01**

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
1.1	Diesel generators and local Control Panels (LCPs)			
1.2	Diesel generators and local Control Panels (LCPs)			
1.3	Un-Interruptible Power Supplies (UPSs)			
2	**Describe the power supplied by the EPG&D systems, to include:**			
2.1	NATO Standard Power (440V, 3 Phase, 60 Hz)			
2.2	Ship's power requirements: 1. 120V, 1 phase, 60 Hz; 2. 220V, 1 phase, 60 Hz; 3. 120V, 3 phase, 400 Hz; and 4. 24V DC			
3	**Describe the categories of power to include**			
3.1	Vital			
3.2	Semi-vital			
5.3	Non-vital power			
4	**Describe casualty power to include:**			
4.1	Sections serviced by casualty power			
4.2	Connection procedures			
4.3	Safety considerations and precautions			
5	**Describe power distribution safety concerns**			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.06 DESCRIBE THE FUNCTION OF MARINE AUXILIARY SYSTEMS

1. Performance. Describe the function of marine auxiliary systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the function of shipboard marine auxiliary systems by describing:

   a. The purpose and function of marine auxiliary systems;

   b. Refrigeration systems;

   c. Compressed gas systems;

   d. Desalination plants;

   e. Steering systems;

   f. Cathodic protection systems; and

   g. Black/grey water sanitary systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the function of marine auxiliary systems, to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**A28,A29,A30,A33,A34,A35,A36,A37,A40**
1.1	The difference between auxiliary and ancillary systems			
1.2	Their contribution to the provision of operations			
2	**Describe the purpose of the refrigeration systems, to include:**			
2.1	Theory of operation			
2.2	Basic cycle components			
2.3	Basic shipboard system			
3	**Describe the purpose of the compressed gas systems, to include:**			
3.1	High Pressure (HP) air			
3.2	Low Pressure (LP) air			
3.3	Contribution to control systems			
3.4	Contribution to weapons systems			
3.5	Safety considerations			
4	**Describe the purpose of the desalination plants, to include:**			

Teaching Points		Instructional Strategies	Time (min)	References
4.1	Physical			
4.2	Thermal			
5	**Describe the purpose of the steering systems, to include:**			
5.1	Hydraulic			
5.2	Electro-hydraulic			
6	**Describe the purpose of the Cathodic Protection Systems to include:**			
6.1	Basic theory			
6.2	Sacrificial protection			
6.3	Impressed current protection			
6.4	Changes in protection requirements			
7	**Describe the purpose of the Black/Grey Water Sanitary System**			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 1 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.07 DESCRIBE HULL SYSTEMS

1. Performance. Describe Hull Systems

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the hull systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Briefing of HVAC system by the CTO/MSEO and/or Senior Hull Technician, to include: 1. Key components; 2. Operational description;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
1.2	Briefing of Main Seawater Services System by the CTO/MSEO/AMSEO and/or Senior Hull Technician, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of HVAC Presentation, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Redundancy and survivability 7. Conclusion			
2.2	Presentation of Main Seawater Services System, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 DESCRIBE THE OPERATION OF MARINE SYSTEM ENGINEERING EQUIPMENT

EO 003.08 DESCRIBE THE MAIN PROPULSION SYSTEM

1. Performance. Describe The Main Propulsion System

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: The candidate shall describe main propulsion system.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, A58, A73, A92, A97**
1.1	Briefing of Main Propulsion Overview by the CTO/MSEO/AMSEO and/or Chief Eng, to include: 1. Key components; 2. Operational description;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
1.2	Briefing of Shaftline by the MSEO and/or Chief Eng, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of Main Propulsion Overview, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			

Teaching Points		Instructional Strategies	Time (min)	References
	7. Conclusion			
2.2	Presentation of Shaftline Overview, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.09 DESCRIBE THE MAIN LUBRICATING OIL SYSTEM

1. Performance. Describe the Main Lubricating Oil Systems

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the main lubricating oil systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Briefing of Main Lubricating Oil System by the CTO/MSEO/AMSEO and/or Chief Eng, to include: 1. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of System(s): Presentation of Main Lubricating Oil System, to include: 1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.10 DESCRIBE THE MACHINERY CONTROL SYSTEM

1. Performance. Describe the Machinery Control System

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the machinery control systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Shipboard Familiarization of Equipment: Briefing of Machinery Control System by the CTO/MSEO/AMSEO and/or Senior IPMS Technician, to include: 1. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of System(s): Presentation of Machinery Control System, to include: 1. System Purpose 2. Key components; 3. Operational description (including Propulsion System Coordinator (PSC) and Automatic Sequence Controller (ASC)); 4. Modes of operation; 5. Control and Monitoring; 6. System Inter-relationships; and 7. Redundancy and survivability 8. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.11 DESCRIBE THE POWER GENERATION AND DISTRIBUTION SYSTEM

1. Performance. Describe the Power Generation and Distribution System

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the power generation and distribution systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Briefing of the Power Generation and Distribution System Electrical Overview by the CTO/MSEO/AMSEO and/or Senior Electrician, to include:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Key components; 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of System(s): Presentation of the Power Generation and Distribution System, to include: 1. System Purpose 2. Key components (including purpose, location, load classifications, and power types); 3. Operational description (including automated functions of the Switchboard Programmable Logic Controllers (PLCs 4. Modes of operation (including system configurations); 5. Control and Monitoring; 6. System Inter-relationships; and 7. Redundancy and survivability			

Teaching Points		Instructional Strategies	Time (min)	References
	8. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.12 DESCRIBE THE STEERING SYSTEM

1. Performance. Describe the Steering System

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the steering systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Briefing of the Steering System (Electrical & Electro-Hydraulic) by the CTO/MSEO/AMSEO and/or Senior Electrician, to include: 1. Key components; 2. Operational description; 3. Modes of operation;			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of System(s): Presentation of Steering System (Electrical & Electro-Hydraulic), to include: 1. Key components(purpose and location); 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – DESCRIBE THE OPERATION OF MARINE SYSTEMS ENGINEERING EQUIPMENT

EO 003.13 DESCRIBE THE FUEL OIL SERVICE SYSTEM

1. Performance. Describe the Fuel Oil Service System

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the fuel oil systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A13, A15, A16, A17, A18, A26, A27, A28, A29, A30, A33, A34, A35, A36, A37, A39, A40, A41, B01, C08, C15, C20**
1.1	Shipboard Familiarization of Equipment, to include a briefing of the Fuel Oil Service System by the CTO/MSEO/AMSEO and/or Main Propulsion Petty Officer: 1. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of System(s): Presentation of Steering System (Electrical & Electro-Hydraulic), to include: 1. Key components(purpose and location); 2. Operational description; 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.01 DESCRIBE WARFARE SYSTEMS REQUIREMENTS

1. Performance. Describe Warfare System Requirements.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the warfare system requirements by describing:

   a. The purpose of a naval warfare system;

   b. The key elements of a warfare system;

   c. Warfare systems onboard RCN major warships; and

   d. The detect to engage sequence.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the purpose of a naval warfare system, including:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**B02,B03,B05,C13,C22**

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Typical platforms: 1. Frigates; 2. Destroyers; 3. AORs; 4. Submarines; and 5. Minor vessels			
1.2	Typical missions assigned to the RCN: 1. Fishery patrol; 2. Drug Interdiction; 3. Land forces support; 4. Convoy protection; and 5. Sovereignty			
1.3	Types of warfare: 1. Anti-Air Warfare (AAW); 2. Anti-Surface Warfare (ASuW); and 3. Anti-Submarine Warfare (ASW)			
2	**Describe the key elements of a warfare system, including:**			
2.1	Sensors			
2.2	Command, control, and communications			
2.3	Weapon			
2.4	Countermeasures			
3	**Describe generic warfare systems**			
3.1	Radar system			
3.2	Sonar system			
3.3	Electronic system			

Teaching Points		Instructional Strategies	Time (min)	References
3.4	Navigation system			
3.5	Command and Control system			
3.6	Communication system			
3.7	Fire control system			
3.8	Weapons system			
4	**Detect to Engage Sequence**			
4.1	Detect			
4.2	Localize			
4.3	Identify			
4.4	Track			
4.5	Engage			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.02 DESCRIBE COMMAND AND CONTROL SYSTEMS (CCS)

1. Performance. Describe Command and Control Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate knowledge of the CCS contribution to a ship's mission by:

   a. Describing the CCS concept;

   b. Drawing a block diagram of the Halifax Class CMS 330 to include:

      (1) Main system components of a CMS 330,

      (2) Systems that interface with a CMS 330.

   c. Defining threat evaluation weapon assignment (TEWA).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the general command and control system concept, to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**100**	**C03,A49**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	Definition and purpose			
1.2	Types of systems			
1.3	Data fusion/correlation			
2	**Draw a generic block diagram, to include:**			
2.1	System components: 1. Processor; 2. Software/storage; 3. Doctrine; 4. Data bus; 5. Human interface; and 6. Interfaces to other systems			
2.2	Interfaced systems: 1. Intelligence; Navigation Suite; 2. EW Suite; 3. Radar Suite; 4. Sonar Suite; and 5. Weapon systems			
3	**Define threat evaluation weapon assignment (TEWA)**			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.03 DESCRIBE NAVAL WEAPONS SYSTEMS

1. Performance. Describe Naval Weapon Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the weapons used in naval warfare systems by describing:

   a. The function of a weapon;

   b. The different types of weapons;

   c. The features;

   d. The fire control principle; and

   e. Drawing a layered defense diagram.

4. Teaching Points:

Teaching Points		**Instructional Strategies**	**Time** (min)	**References**
1	**Define the function of a weapon**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	100	**C13**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe the different types of weapons, to include:**			
2.1	Projectiles			
2.2	Missiles			
2.3	Torpedoes			
2.4	Bombs			
3	**Describe general features, to include:**			
3.1	Guidance			
3.2	Propulsion(Impulse, rocket)			
3.3	Warhead			
3	**Define the fire control principle, to include:**			
4.1	Fire control problem			
4.2	Fire control solution: 1. Moving platform; 2. Moving target; 3. Moving medium; and 4. Moving projectile			
5	**Draw a basic layered defence diagram, to include short, medium, and long range weapons and sensors**			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.04 DESCRIBE THE RADAR SYSTEMS

1. Performance. Describe the Radar Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the radar systems by :

   a. Defining radio detection and ranging;

   b. Drawing the basic operating principle of a RADAR;

   c. Describing the basic operating principle of a RADAR;

   d. Listing RADAR types; and

   e. Drawing RADAR block diagrams.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define Radio Detection And Ranging**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**100**	**A1,A2,C17**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe and draw the basic operation principles of RADAR systems**			
3	**List the target information retrieve from a RADAR echo, to include:**			
3.1	Range			
3.2	Bearing			
3.3	Elevation			
3.4	Doppler			
4	**List RADAR types, to include:**			
4.1	Navigation			
4.2	Air Search			
4.3	Surface Search			
4.4	Fire Control			
4.5	Secondary(IFF)			
5	**Give an example of each radar type above as fitted on whichever platform will be used for the NEI Afloat phase**			
6	**Draw basic RADAR block diagram, to include:**			
6.1	Signal Processor			
6.2	Transmitter			
6.3	Duplexer			
6.4	Antenna system			
6.5	Receiver			
6.6	Display			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout; and

   b. Multimedia classroom.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.05 DESCRIBE SONAR SYSTEMS

1. Performance. Describe the Sonar Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the SONAR systems by:

   a. Defining SONAR;

   b. Drawing the operating principle of SONAR;

   c. Describing the operating principle of SONAR;

   d. Listing SONAR types; and

   e. Drawing a SONAR block diagram.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define Sound Navigation And Ranging**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint	**100**	**C04, C16**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2	**Describe and draw the operating principles of an active SONAR and a passive SONAR system**			
3	**List the target information received from active and passive SONARs, to include:**			
3.1	Range			
3.2	Bearing			
3.3	Doppler			
3.4	Identity			
4	**List SONAR types, to include:**			
4.1	Submarine detection			
4.2	Mine detection			
4.3	Echo sounding			
5	**Give an example of each SONAR type above as fitted on whichever platform will be used for the NEI Afloat phase**			
6	**Draw basic active SONAR block diagram, to include:**			
6.1	Synchronizer			
6.2	Transmitter			
6.3	Duplexer			
6.4	Acoustic array			
6.5	Receiver			
6.6	Display			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.06 DESCRIBE NAVIGATION SYSTEMS

1. Performance. Describe the Navigation Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the navigation systems by:

   a. Defining the function of navigation systems;

   b. Describing navigation equipment; and

   c. Drawing navigation systems diagram.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define the function of a navigation system**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**A2,A14,A44,A45,A47,A48,A51,A68**

Teaching Points		Instructional Strategies	Time (min)	References
2	**List the information provided by a navigation system, to include:**			
2.1	Heading			
2.2	Roll			
2.3	Pitch			
2.4	Speed			
2.5	Position			
3	**Describe navigation equipment, to include:**			
3.1	INS			
3.2	GPS			
3.3	Doppler speed log			
3.4	Depth Sounder			
3.5	SHipboard INtegrated NAvigation and Display System (SHINNADS)			
3.6	Navigation Data Distribution System (NDDS)			
3.7	Meteorological Systems			
4	**Draw a basic navigation system diagram, to include:**			
4.1	Navigation equipment			
4.2	INS			
4.3	GPS			
4.4	Doppler Speed log			
4.5	Depth Sounder			
4.6	Navigation Data Distribution System (NDDS)			
4.7	End users, to include: 1. SHipboard INtegrated NAvigation and Display System (SHINNADS); 2. Sensors; 3. Weapons; and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. CCS			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.07 DESCRIBE NAVAL COMMUNICATION SYSTEMS

1. Performance. Describe Naval Communication Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of thenaval communication system by:

   a. Defining the function of a communication system;

   b. Describing the components of a communication system;

   c. Describing the requirements for naval communications;

   d. Describing communication systems used onboard RCN major warships.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define the function of a communications system**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**A42,A64,C12**

Teaching Points		Instructional Strategies	Time (min)	References
2	**List the main types of communications onboard a ship, to include:**			
2.1	Internal communications			
2.2	External communications			
3	**Describe the components of a communication system, to include:**			
3.1	message, to include: 1. Voice; and 2. Data.			
3.2	transducer, to include: 1. energy conversion; and 2. baseband signal			
3.3	transmitter, to include: 1. Digital/Analogue conversion 2. Modulation Amplifies			
3.4	channel, to include: 1. Types: 1.1. atmosphere/space; 1.2. telephone cables; 1.3. Water; and 1.4. Optical fiber. 2. Noise: 2.1. Atmosphere; and 2.2. Circuitry noise			
3.5	Receiver			
3.6	Output message			
4	**List the requirements for naval communications, to include:**			
4.1	Freedom of motion			

Teaching Points		Instructional Strategies	Time (min)	References
4.2	Reliability			
4.3	Interoperability			
4.4	Security			
5	**List and describe RCN communications systems to include**			
5.1	SATCOM			
5.2	LINK			
5.3	GCCS			
5.4	Radio			
5.5	SHINCOM			
5.6	Sound powered communications (SOCOM)			
6	**List and describe the between communications systems, to include:**			
6.1	SHINCOM terminals			
6.2	High Speed Data Connectivity (HSDC)			
6.3	Public address system			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period; and

   b. 0 x 50 min periods – EC

6. Training and Learning Aids:

   a. Student handout.; and

   b. Multimedia Classroom

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.08 DESCRIBE ELECTRONIC WARFARE (EW) SYSTEMS

1. Performance. Describe Electronic Warfare Systems.

2. Conditions:

   a. Given:

      (1) Nil.

   b. Denied:

      (1) References; and

      (2) Assistance.

   c. Environment:

      (1) Classroom.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the EW systems by:

   a. Defining the purpose of Electronic Warfare;

   b. Describing the functions of an Electronic Support (ES) System;

   c. Describing the functions of an Electronic Attack (EA) System; and

   d. Describing Electronic Protection (EP) techniques.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define the purpose of Electronic Warfare; and List and describe the components of EW, to include:**	**Method:** Interactive Lecture / discussion **Media:** PowerPoint **Environment:** Classroom	**100**	**C13**

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Electronic Support(ES)			
1.2	Electronic Attack (EA)			
1.3	Electronic Protection (EP)			
2	**Describe the functions of an ES system, to include:**			
2.1	Radar warning receiver			
2.2	Communications intercept			
2.3	Emitter location			
2.4	Identification			
2.5	Electronic intelligence gathering			
3	**Describe the functions of an EA system, to include:**			
3.1	Electronic Jamming			
3.2	Chaff			
3.3	Flares			
3.4	Active Decoys			
3.5	Signature Reduction			
3.6	EMCON			
4	**Describe EP techniques, to include:**			
4.1	Frequency Hopping			
4.2	Range/Doppler gates			
4.3	Sidelobe suppression			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids:

   a. Student handout.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.09 DESCRIBE THE COMMAND AND CONTROL SYSTEMS

1. Performance. Describe the Command and Control Systems.

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the command and control systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing of the Command and Control System by the CTO/CSE and/or A/CSEO, to include: 1. Key components; 2. Functional description (data paths and flow);			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Command and Control System, to include: 1. System Purpose 2. Key components (purpose and location); 3. Functional description (data flow and usage during engagement; 4. Modes of operation; 5. System Inter-relationships; and 6. Redundancy and survivability 7. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.10 DESCRIBE THE WEAPON AND SENSOR ENVELOPES

1. Performance. Describe the Weapon and Sensor Envelopes

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the command and control systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing of the Ship's Weapons and Sensors by the CTO/CSEO and/or A/CSEO, to include: 1. Key components; 2. Weapons' Specifications;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Sensors' Specifications; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Command and Control System, to include: 1. System Purpose 2. Concept of Layered Defense; 3. Guidance types; 4. Organic and Inorganic sensors; 5. Specifications, capabilities and limitations of AAW, ASuW and ASW weapons; 6. Respective Firing Locations for weapons (Primary, Alternate, secondary and emergency (as applicable)); 7. Specifications, capabilities and limitations of AAW, ASuW and ASW sensors; 8. Key components (purpose and location); 9. System Inter-relationships; and			

Teaching Points		Instructional Strategies	Time (min)	References
	10. Redundancy and survivability 11. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.11 DESCRIBE THE NAVIGATION SUITE

1. Performance. Describe the Navigation Suite

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the navigation suite.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing of Ship's Navigation Systems and Interfaces by the CTO/CSEO/ACSEO to include: 1. Purpose of Navigation Systems 2. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Functional description (data paths and flow); 4. Modes of operation; 5. Control and Monitoring; 6. System Inter-relationships; and 7. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of Ship's Navigation Systems and Interfaces, to include: 8. System Purpose 9. Key components(purpose and location); 10. Functional description (data flow and usage during engagement; 11. Modes of operation; 12. System Inter-relationships; and 13. Redundancy and survivability 14. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.12 DESCRIBE THE RADAR SUITE

1. Performance. Describe the Radar Suite

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the radar suite.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing of the Ship's Radar Suite by the CTO/CSEO and/or A/CSEO, to include: 1. Key components; 2. Functional description (data paths and flow);			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Ship's Radar Suite, to include: 1. System Purpose 2. Generic Monostatic Radar Components Operation 3. Key components(purpose and location); 4. Functional description (data flow and usage during engagement; 5. Modes of operation; 6. System Inter-relationships; and 7. Redundancy and survivability 8. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.13 DESCRIBE THE INTERNAL AND EXTERNAL COMMUNICATION SYSTEM

1. Performance. Describe the Internal and External Communication systems

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the internal and external communication systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing by CTO/CSEO/ACSEO of the Ship's Internal and External Communications Systems to include: 1. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Functional description (data paths and flow); 3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Internal and External Communications Systems, to include: 1. System Purpose 2. Key components(purpose and location); 3. Operational description; 4. Modes of operation; 5. Modulation 6. Control and Monitoring 7. Use of a Communication Plan (COMPLAN) 8. System Inter-relationships; 9. Redundancy and survivability 10. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.14 DESCRIBE THE ELECTRONIC WARFARE SUITE

1. Performance. Describe the Electronic Warfare Suite

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the electronic warfare suite.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing by CTO/CSEO/ACSEO of the Ship's Electronic Warfare Suite to include: 1. Key components; 2. Functional description (data paths and flow);			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Modes of operation; 4. Control and Monitoring; 5. System Inter-relationships; and 6. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Electronic Warfare Suite, to include: 1. System Purpose 2. Key components(purpose and location); 3. Operational description; 4. Modes of operation; 5. Electronic Protection, Electronic Attack and Electronic Support; 6. Control and Monitoring 7. System Inter-relationships; and 8. Redundancy and survivability 9. Conclusion			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – DESCRIBE THE OPERATION OF COMBAT SYSTEMS ENGINEERING EQUIPMENT

EO 004.15 DESCRIBE THE FIRE CONTROL SYSTEM

1. Performance. Describe the Fire control system

2. Conditions:

   a. Given:

      (1) Operational ship;

      (2) Supervision;

      (3) Assistance; and

      (4) References.

   b. Denied: Nil.

   c. Environment: At sea or alongside in any condition day or night.

3. Standard: IAW the references listed in Para 4, the trainee shall demonstrate technical knowledge of the fire control system.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard Familiarization of Equipment, to include:**	**Method:** OJT **Media:** N/A **Environment**: at sea or alongside	**Throughout the afloat phase**	**A1, A2, A14, A42, A44, A45, C03, A47, A48, A49, C04, A51, A64, B02, A68, B03, B05, C12, C13, C16, C17, C22**
1.1	Briefing by CTO/CSEO/ACSEO of the Ship's Fire Control System to include: 1. System Purpose 2. Key components;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Functional description (data paths and flow); 4. Modes of operation; 5. Control and Monitoring; 6. System Inter-relationships; and 7. Redundancy and survivability			
2	**Presentation of Systems**			
2.1	Presentation of the Fire Control System, to include: 1. System Purpose 2. Definition of the Fire Control Problem 3. Key components(purpose and location); 4. Operational description; 5. Modes of operation; 6. Control and Monitoring 7. System Inter-relationships; 8. Redundancy and survivability; 9. Conclusion.			

5. Total EO Time: Total: 0 x 50 minute periods as follows:

   a. 0 x 50 min period;

   b. 0 x 50 min periods – EC;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	1/1555-UAZ10251/20 en A	System Description SG180
A02	3749DOC020102	NSC Radar Operator Manual
A03	9800K002D033	HCM Electronic Support Measures
A04	A-AE-000-001/AG-E00	Guide to the Divisional System (RCN, 2014)
A05	A-PD-055-002/PP-001	Canadian Armed Forces Military Employment Structure (MES), Volume 2 - Occupational Specifications, Part 1 - Officer Occupation Specifications
A06	A-PD-055-002/PP-002,	Canadian Armed Forces Military Employment Structure (MES), Volume 2 - Occupational Specifications, Part 2 - Non-Commissioned Member Occupation Specifications
A07	A-PD-055-002/PP-003	Canadian Forces Officer General Specification (OGS)
A08	A-PD-055-002/PP-004,	Canadian Armed Forces Military Employment Structure (MES), Volume 2 - General Specifications, Part 4 - Non-Commissioned Member General Specifications
A09	B-GJ-005-000-FP-001	CJFP 01 - Canadian Military Doctrine
A12	C-03-005-012/AM-001	Naval Materiel Management System Manual
A13	C-03-005-033/AA-000	Naval Engineering Manual
A14	C-21-132-000/MS-001	Meteorological Station AN/SMH-502 Technical Manual, 1995-10-27
A15	C-23-045-000/AG-001	Damage Control (Sea) Volume 1
A16	C-23-HFX-000/MB-001	Propulsion System Operating Manual
A17	C-23-HFX-000/MS-003	Manual of Trim and Stability Applicable to Halifax Class
A18	C-24-293-000/MS-001	Main Propulsion System Manual for Halifax Class Ships
A19	C-24-545-000/MS-001	Stern and Bulkhead Seals
A20	C-24-545-000-MS-001	Mane Type Shaft and Bulkhead Seal
A21	C-24-545-A00/MS-002	CPF Main Shaft Flexible Coupling
A22	C-24-545-A00-MS-005	Main Shaft Flexible Coupling
A23	C-24-545-B00/MS-001	Thrust and Plummer Bearing
A24	C-24-559-000-MS-001	CRPP and Shafting
A25	C-25-509-000/MS-001	Fuel Oil Service
A26	C-26-320-000/MS001	Primary Electrical Power Generation and Distribution System Manual Applicable to Halifax Class
A27	C-26-325-000-MS-001	Casualty Power System Components
A28	C-27-778-000-MS-001	Steering System
A29	C-27-783-000-MS-001	HP Air System
A30	C-27-784-000-MS-001	LP Air System
A31	C-27-791-000/MP-001	Seawater Services
A32	C-27-876-000/MP-001	Diesel Driven Fire Pump
A33	C-27-890-000-MS-002	Sewage Treatment Plant
A34	C-27-948-000-MS-001	Reverse Osmosis Desalination System

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A35	C-27-952-000-MS-001	Vacuum Sewage Collection System
A36	C-29-354-000-MS-001	Main Refrigeration System Manual
A37	C-29-355-000/MS-001	Heating Ventilation and Air Conditioning System
A38	C-03-020-000/NT-001	Trial Agenda for Dynamic Sound Ranging – Applicable to Halifax Class
A39	C-38-224-000/MS001	Secondary Electrical Power Distribution and Lighting System Applicable to Halifax Class
A40	C-38-231-001-MS-001	Cathodic Protection (CAPAC) System
A41	C-48-102-000/MD-001	Equipment Description Integrated Machinery Control System (IMCS) for Halifax Class
A42	C-51-501-000/MS-001	External Naval Communications System Applicable to Halifax Class
A43	C-51-575-000/MS001	Communications Control and Monitoring System
A44	C-57-574-000/MS-001	Navigation and Weatherdeck Lighting Control Equipment, Ch/Mod. 2 1998-10-16
A45	C-57-611-000/CS-001	SRD-331 Doppler Speed Log System
A47	C-57-633-A00/MS-001	Remote Display Inertial Navigation System AN/SYN-502 (Applicable to HALIFAX Class), 1993-05-10
A48	C-57-633-C00/MS-001	Ships Course Indicator Operating, Troubleshooting, Maintenance, Inspection and Illustrated Parts Breakdown,
A49	DAB-DOC-002	Maintainer Technical Documentation Command and Control System Combat System Computing Environment
A51	C-57-787-000/MS-001	Shipboard Integrated Navigation and Display System (SHINNADS)
A52	C-58-259-000/MD-001	Theory of Operation of Countermeasures Set AN/SLQ-503
A53	C-58-393-000/MS001	Operation and Maintenance Instructions - Direction Finder Set AN/ARD-504
A54	C-59-764-002/MB-001	SPSS 3-1 Sonobuoy Processor Subsystem Technical Manual
A55	C-69-703-000/MS-001	Sonar Hull Outfit C5 (HOC5) Operating and Maintenance Instructions with Parts List (Applicable to HFX and IRO Classes)
A56	C-70-377-000/MD001	Equipment Description and Operation Launching System Guided Missile Mk 48 Mod 4 Applicable to Halifax Class
A57	CANACCS9LV-SHM63-SD001-00	SAAB FCS System Description
A60	CANFORGEN 218/12 CMP 101/12 011212Z Nov 12	Canadian Forces Junior Officer Development (CFJOD) Programme
A61	CFAO 43-7	Captain's Ship's Book. http://corpsec.mil.ca/admfincs/subjects/cfao/043-07_e.asp
A62	CFCD 106	
A63	CFCD 109	Naval Vessel Signature Control Manual. http://halifax.mil.ca/NFR/pages/NFR_UWW.html

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A64		CPF SHINCOM 3100 Technical Manual MAN-1148, Rev B
A65	D-03-002-000/SG-006	Habitability and Husbandry Standard for Canadian Forces Ships
A66	D-03-002-008/SG-011	Requirements for docking/undocking of CF Vessels
A68	E-57-502-000	Halifax Class Navigation System Group, LCdr Ouellet DMSS 8-5. Rev 1.2.0, 27 Apr 12.
A74		LCdr Bradford, "Naval Technical Officers Career Manager Brief." http://cmpapp.mil.ca/dgmc/en/career/briefing/00346.ppt
A75		LCdr Williams, "Naval Technical Officers Career Manager Brief." http://cmpapp.mil.ca/dgmc/en/career/briefing/00344-00345.ppt
A80		Marine Systems Engineering Officer. Government of Canada, National Defence, ADM(PA), DGM (Director General, Marketing), DEC (Directorate Electronic Communications). Canadian Armed Forces Recruiting, n.d. Web. 06 June 2014. http://www.forces.ca/en/job/marinesystemsengineeringofficer-84
A81		Maritime Surface and Sub-surface Officer. Government of Canada, National Defence, ADM(PA), DGM (Director General, Marketing), DEC (Directorate Electronic Communications). Canadian Armed Forces Recruiting, n.d. Web. 09 June 2014. http://www.forces.ca/en/job/maritimesurfaceandsubsurfaceofficer-65
A83		Naval Combat Systems Engineering Officer. Government of Canada, National Defence, ADM(PA), DGM (Director General, Marketing), DEC (Directorate Electronic Communications). Canadian Armed Forces Recruiting, n.d. Web. 06 June 2014. http://www.forces.ca/en/job/navalcombatsystemsengineeringofficer-82
A85	NAVGEN 017/15 081834Z SEP 15	http://marcom-comar.mil.ca/navgens/navgens-eng.asp?yr=2015
A86	NAVORD 1000-0	Naval Order – Policy
A87	NAVORD 3395-1	Naval Signature Management Policy and Organization. http://marcom-comar.mil.ca/repository/navords/3000/3395-1-eng.pdf
A98		Ship’s Standing Orders AL 11. http://halifax.mil.ca/SEA_TRG/pages/references.html

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		Canadian Occupational Health and Safety Regulations, Part VIII, Electrical Safety. http://laws.justice.gc.ca/eng/regulations/sor-86-304/page-28.html#h-81
B02		Destroyer Class Page. http://www.navy-marine.forces.gc.ca/en/fleet-units/destroyers-home.page
B03		Frigate Class Page. http://www.navy-marine.forces.gc.ca/en/fleet-units/frigates-home.page
B04		Learning and Career Centres (LCCs). Human Resources-Civilian (HR-Civ) Intranet. Assistant Deputy Minister Human Resources, 08 May 2014. http://hrciv-rhciv.mil.ca/en/p-learning-lcc-halifax.page
B05		MCDV Class Page. http://www.navy-marine.forces.gc.ca/en/fleet-units/mcdv-home.page
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		Boutilier, James A. The RCN in Retrospect, 1910-1968. Vancouver: University of British Columbia, 1982.
C02		Bruce, A. P. C. "An Encyclopedia of Naval History." London: Thistle, 2014. 164.
C03		Naeem, H. et al. A Semi-Automated Approach to Threat Evaluation and Optimal Defensive Resource Allocation. Springer-Verlag Berlin Heidelberg, 2009.
C04		Underwater Acoustic System Analysis; Burdic, William S.; Prentice-Hall, 1984.
C05		Gray, D. “Introduction. Centralized and Automatic Controls in Ships.” Oxford: Pergamon, 1966
C06		Griffiths, D. "British Marine Industry and the Diesel Engine." The Northern Mariner 7.
C07		Higgitt, M., "Through Fire and Water, HMS Ardent: The Forgotten Frigate of the Falklands War," Mainstream Publishing Company, 2001.
C08		IPMS Training Manual. Gas Tops, Ltd. Lisa Cornfield, P.Eng. April 2002
C09		Legion Magazine, "On This Date - December 2010 - 3 December 1969." December 1, 2010.
C10		Levinson, J.L. and Edwards, R.L., "Missile Inbound", Annapolis: Naval Institute Press, 1997.
C11		Lyon, D. and Winfield, R. “The Sail & Steam Navy List: All the Ships of the Royal Navy, 1815-1889.” London: Chatham, 2004.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
C12		Naval Shipboard Communications Systems - John C. Kim, Eugen I. Muehldorf
C13		Payne, C. Principles of Naval Weapon Systems. 2nd ed. U.S. Naval Institute Press. 2010.
C14		Phillips, R. "A Damn Close-Run Thing: A Brief History of the Falklands Conflict." Shilka, 2011.
C15		Pike, John. "The Hull." Globalsecurity.org, 07 July 2011. Web. 09 June 2014. http://www.globalsecurity.org/military/systems/ship/hull.htm
C16		Principles of Underwater Sound, 3rd ed.; Urick, Robert J.; McGraw-Hill, 1983.
C17		Radar Fundamentals; Faulconbridge, R. Ian; Argos Press (Scitech Publishing) 2002.
C18		Rogers, W. and Rogers, S. "Storm Center: The USS Vincennes and Iran Air Flight 655: A Personal Account of Tragedy and Terrorism." Annapolis: Naval Institute Press, 1992.
C19		Saarlas, M. “Steam and Gas Turbines for Marine Propulsion.” Annapolis, MD: Naval Institute, 1978.
C20		Scherer, T. The Evolution of Machinery Control Systems Support At the Naval Ship Systems Engineering Station. 2011. https://seagrant.mit.edu/ESRDC_library/Scherer-Evolution-ASNE-2011.pdf
C21		Steamboat Bill: Journal of the Steamship Historical Society of America 50 (1993): 56.
C22		Submarine Class Page. http://www.navy-marine.forces.gc.ca/en/fleet-units/submarines-home.page
C23		Tremblay, C. "HMCS Kootenay - 40 Years of Lessons Learned." Maritime Engineering Journal No. 65 Fall 2009 / Winter 2010.
D		**Other Publications and Manuals that relate to this Qualification**
D01		Canadian sailor dies in submarine accident. CBC News. 7 October 2004. http://www.cbc.ca/news/canada/canadian-sailor-dies-in-submarine-accident-1.478149
D02		Navy Reverting to Steel in Shipbuilding after Cracks in Aluminum. The New York Times. Associated Press. 11 August 1987. http://www.nytimes.com/1987/08/11/us/navy-reverting-to-steel-in-shipbuilding-after-cracks-in-aluminum.html
D03		Nine hurt on stricken submarine. BBC News. 6 October 2004. http://news.bbc.co.uk/2/hi/uk_news/scotland/3717906.stm

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0005	Defence Services Program (DSP)	Train	002	
K0010	Technical Directives and Instructions	Train	002	
K0015	Role and employment of the MARE in the Canadian Navy	Train	001	
K0020	Career progression of the MARE	Train	001	
K0025	Characteristics and effects of shipboard weapons systems	Train	004	
K0030	Operational Readiness criteria	Train	002	
K0035	Effects of damage on operational effectiveness and ship safety	Train	002	
K0040	Ship's technical readiness	Train	002	
K0045	Ship's capabilities and limitations in various	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
	Damage Control/NBC conditions			
K0050	Ship's weight control policy	Train	002	
K0065	Duties of the Engineering Watch	Train	001	
K0070	Capabilities and limitations of ship systems under limited/damaged operating conditions	Train	002	
K0075	Shipboard emergency power distribution	Train	003	
K0080	Ship radiological protection systems	Train	003	
K0085	Emergency Response Team Package (ERT PACK)	Train	002	
K0090	Combat systems engineering emergency procedures	Train	002	
K0095	Safe handling and storage of explosive material onboard ship	Train	003,004	
K0100	Shipboard main armament safety precautions	Train	003	
K0110	Principles of safety of explosives, fuses, detonation systems, Pyrotechnics, and propellants of all ammunition	Train	003,004	
K0115	Identification of shipboard ammunition	Train	002,004	
K0120	Handling and stowage of ammunition and explosives	Train	004	
K0125	Otto (torpedo) fuel safety procedures	Train	002	
K0135	Ammunition Allotment/Expenditure guidelines (MCAAL)	Train	003	
K0140	Ship and systems layout	Train	002,003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0145	Purpose of ship structure and structural components	Train	002,003	
K0150	Principles of ship stability	Train	002,003	
K0155	DND drawing policy, practices and acceptances procedures.	Train	002,003	
K0160	Ship Systems Engineering (SSE) and ship level integration of the hull, marine and combat systems	Train	002,003	
K0165	Shipboard habitability considerations	Train	002,003	
K0170	Shipboard noise generation	Train	002	
K0175	Principles of signature reduction	Train	002	
K0180	Recognition of potential "noise shorts"	Train	002	
K0185	Acoustic and thermal functions/properties of shipboard insulation	Train	002	
K0190	Materials used in the marine environment	Train	002,003	
K0195	Processes and principles of corrosion	Train	002,003	
K0200	Ship's datums and levels	Train	002,003	
K0205	The effects of nuclear radiation and Electromagnetic Pulse (EMP)	Train	002,004	
K0210	Equipment Health Monitoring (EHM)	Train	002,003	
K0215	Special tools and test equipment (to include ATE/BITE)	Train	002,003	
K0220	Non Destructive Examination (NDE)	Train	002,003	
K0225	Vibration Analysis (VA)	Train	002	
K0230	Electrical engineering principles	Train	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0235	Mechanical engineering principles	Train	003,004	
K0240	Heat transfer principles	Train	002,003	
K0245	Fluid dynamics principles	Train	002,003	
K0250	Computer engineering principles	Train	002,003	
K0255	Instrumentation Principles	Train	002,003	
K0260	Naval Engineering Manuals (NEM)	Train	002,003	
K0265	Engineering and Test Establishments	Train	003	
K0270	Defence Research Establishments (DRE)	Train	003	
K0275	International Standards Organization (ISO) 9000 Quality Assurance standards	Train	003	
K0280	DND Quality Assurance policies, standards, records, reports / contractor's inspection and quality systems	Train	003	
K0285	Thermodynamic analyses	Train	002,003	
K0290	Hydraulics Theory	Train	002,003	
K0295	Control theory	Train	003	
K0300	Oil and Coolant Condition Analysis Program (OCCAP)	Train	002	
K0310	Technical Statement of Requirements (TSORs)	Train	002	
K0315	DND contract practices and procedures	Train	002	
K0320	Inclining experiment	Train	003	
K0325	Statement of Work (SOW)	Train	003	
K0330	Canadian Standards Association (CSA)	Train	002	
K0340	Military standards (DODS, MILSTDS, NES)	Train	003	
K0345	Shipboard lifting appliances	Train	003	
K0350	Principles of gearing	Train	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0355	Principles of pumps	Train	003	
K0360	Physical properties of lubricating oils	Train	002,003	
K0365	Fastners and shocking of mechanical equipment	Train	002	
K0370	Environmental effects (internal & external) on ships systems	Train	002	
K0380	Principles and types of propulsion systems	Train	003	
K0390	MARCORD 36-50 (Environmental Protection Management)	Train	003	
K0395	MARCORD 36-55 (HAZMAT)	Train	003	
K0400	Canadian Environmental Assessment Act - Impact analysis requirement	Train	003	
K0405	Fisheries Act - Pollution Prevention, Sections 34-42.1	Train	003	
K0410	Canadian Environmental Protection Act (CEPA) Part 6 -Ocean Dumping	Train	002	
K0415	MARPOL regulations (international standards for waste disposal)	Train	002	
K0425	Production industry performance measurement techniques	Train	003	
K0430	Reliability, Availability and Maintainability (RAM)	Train	002	
K0435	Failure mechanisms of materials	Train	003	
K0445	Maintenance levels (first line, second line, third line)	Train	002	
K0450	Shipboard insulation installation considerations	Train	002	
K0460	Characteristics, capabilities, and limitations of Canadian Naval	Train	003,004	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
	ammunition and explosives			
K0465	Human engineering factors in ship design	Train	002	
K0470	Design considerations in ship's pipe work	Train	003	
K0475	MARCOM organization, procedures and policies for the maintenance support of ships	Train	002,003	
K0480	Capabilities, and limitations of DND repair and engineering support organizations	Train	002,003	
K0485	Departmental maintenance requirements	Train	002,003	
K0490	Maintenance Management Information Systems	Train	002	
K0495	Ship's Maintenance Management Information System Manual (SMMISM) (C-03-005-012/AM-002)	Train	002	
K0500	Canadian Forces Supply System (CFSS)	Train	002,003	
K0510	Shipboard Fitted Equipment List (FEL)	Train	002	
K0515	Capabilities and limitations of ships on-board repair facilities	Train	002,003	
K0520	Responsibilities of shore authorities and ship's officers during docking/undocking	Train	002	
K0525	Naval Equipment Index (NEI)	Train	002	
K0530	Shipyard and naval dockyard operations	Train	002	
K0535	Formation Technical Authority	Train	002	
K0540	Work period maintenance program	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0545	Underwater repair/maintenance of ship's systems	Train	002	
K0550	Fleet Maintenance Facility	Train	002	
K0555	Shipboard Non-Destructive Examination (NDE) requirements	Train	002	
K0560	Organization of Canadian Forces Armament Depots (CFADs)	Train	002,004	
K0565	Fleet Diving Unit (FDU) capabilities	Train	002	
K0580	Paint/coating systems	Train	002	
K0595	Policies and procedures for tests and trials	Train	002	
K0600	Procedures for submission, approval and promulgation of CF Naval O&M policies	Train	002	
K0605	Concept, capabilities and use of Naval Maintenance Management System (NAMMS)	Train	002	
K0615	Policies and procedures for Life Cycle Materiel Management (LCMM)	Train	002	
K0625	Equipment Support List (ESL)	Train	002	
K0630	Acquisition of spares	Train	002	
K0635	Cost estimating procedures	Train	002	
K0650	Concepts and principles of life cycle costing (LCC)	Train	002	
K0655	Process for reviewing and amending engineering/technical specifications	Train	002	
K0660	Engineering Change Specifications (ECP)	Train	002	
K0665	Policies and procedures for configuration changes	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0670	Weight control process in HMC ships	Train	002	
K0675	Engineering Change Proposal process	Train	002	
K0680	Submarine safe-to-dive certification	Train	002	
K0690	DND procurement policy	Train	002	
K0695	Field Service Representative (FSR) tasking	Train	002	
K0700	CF Quality Assurance organization	Train	002	
K0705	Statement of Capability Deficiency (SCD)	Train	002	
K0710	Logistics Support Analysis (LSA)	Train	002	
K0715	Dynamic Deliverable List (DDL)	Train	002	
K0720	Project Management	Train	002	
K0725	Integrated Logistics Support (ILS)	Train	002	
K0740	Management and scheduling techniques (e.g. Network Analysis, Critical Path, PERT, etc.)	Train	002	
K0745	Risk analysis	Train	002	
K0750	Proposal evaluation	Train	002	
K0755	Reliability Centered Maintenance (RCM)	Train	002	
K0785	Combat Systems technologies	Train	004	
K0790	Theory and operation of navigation systems	Train	004	
K0795	Gyro Systems	Train	002,004	
K0800	Signal processing techniques	Train	004	
K0805	Naval Tactical Data Systems (NTDS)	Train	002,004	
K0810	Weapon/sensor integration	Train	003,004	
K0815	Threat Evaluation Weapon Assignment (TEWA)	Train	003,004	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0820	Display systems	Train	003	
K0825	Tactical Information Processing Systems (AN/UYK 502, 505, 507, MRM)	Train	003,004	
K0830	Combat Systems Alignment	Train	003	
K0835	Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC) control	Train	004	
K0840	Network concepts applicable to naval information processing systems	Train	003,004	
K0845	Capabilities and limitations of the IFF/SIF equipment	Train	003,004	
K0850	Detection theory for communication systems	Train	004	
K0855	Communications systems and equipment	Train	003,004	
K0860	Frequency Spectrum Management	Train	002,003	
K0865	Communications Security	Train	003,004	
K0870	Data Link Systems	Train	003,004	
K0875	Ship's internal communication systems	Train	004	
K0880	Ship's sensor systems, associated processing and display sub-systems and equipment	Train	003,004	
K0885	Target Motion Analysis (TMA)	Train	003,004	
K0890	Multi-sensor correlation	Train	003,004	
K0895	Automated Detection and Tracking Systems	Train	003,004	
K0900	Software design/development requirement process	Train	003	
K0905	Software Configuration Management	Train	002,003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K0910	Computer technology including principles of data transfer, data communications, and host-target systems relationships	Train	003	
K0915	Software engineering principles	Train	003	
K0920	Oceanographic principles	Train	003,004	
K0925	Underwater acoustics	Train	003,004	
K0930	Theory and operation of sonar systems	Train	003,004	
K0935	Construction and operation of transducers and arrays	Train	003,004	
K0940	Electro-acoustic transducer design and theory	Train	003,004	
K0945	Passive sonar systems	Train	003,004	
K0950	Active Sonar Systems	Train	003,004	
K0955	Sonar Performance Figure (SPF)	Train	002,003	
K0960	Sound Range Trials	Train	002	
K0965	Hull Outfits	Train	003	
K0970	Towed Bodies	Train	003	
K0975	Acoustic Range Prediction System (ARPS)	Train	003	
K0980	Non-Sonar Acoustic equipment	Train	004	
K0985	Torpedo systems used in the Canadian Navy	Train	004	
K0990	Capabilities and limitations and operating characteristics of ASW torpedoes	Train	004	
K0995	Acoustic counter measures	Train	004	
K1005	Close In Weapons System (CIWS)	Train	002,004	
K1010	Gun Weapons systems	Train	004	
K1015	Misfire and hot gun procedures	Train	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K1020	Principles/theory of anti-air (AA) warfare	Train	002	
K1025	Missile systems	Train	004	
K1030	Elements and characteristics of missile warheads (payload, fusing, safety and arming, target kill parameters)	Train	002,004	
K1035	Preparation for missile firing on HMC ships	Train	002,004	
K1040	Ballistics	Train	002,004	
K1045	Countermeasures against missile threat	Train	002	
K1050	Characteristics, capabilities and limitations of surface, AA targets and target towing vehicles	Train	003	
K1055	Weapons firing record-keeping	Train	002	
K1060	Electronic warfare (EW) theory	Train	003,004	
K1065	EW systems	Train	002,004	
K1070	Radar Theory	Train	002,004	
K1075	TACAN and radio beacons	Train	003,004	
K1080	Navigation/surveillance Radars	Train	002,004	
K1085	Fire Control Radars/CWI	Train	004	
K1090	Electro-Optic Systems (e.g. - IR devices, lasers, Low-Light level TV)	Train	002,004	
K1095	Telemetry	Train	002	
K1100	Ship survivability, including susceptibility and vulnerability	Train	002	
K1105	Ship design concepts to reduce vulnerability	Train	002	
K1110	Shipboard radar cross section	Train	002	
K1115	Noise attenuation arrangements	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K1120	Shipboard effects of shock, vibration or blast	Train	002,003	
K1125	Marine and Combat Systems and there ship level design impact	Train	002	
K1130	Ship signature reduction techniques	Train	002	
K1135	Ship motion theory	Train	003	
K1140	Canadian Forces Technical Orders (CFTOs)	Train	002	
K1145	Maintenance administration in HMC ships	Train	002	
K1150	Naval maintenance documentation	Train	002	
K1155	Routine engineering reports promulgated by HMC ships	Train	002	
K1160	Operational Deficiency (OPDEF)	Train	002	
K1165	Unsatisfactory Condition Report (UCR) procedures	Train	002	
K1170	Pre-Installation Failure (PIF) Report procedures	Train	002	
K1175	Engineering Officers Technical Instructions and memorandums	Train	002	
K1180	Periodic Engineering Letter (PEL)	Train	002	
K1190	NDHQ Engineering Support Organization	Train	002	
K1210	Fleet Support planning	Train	002	
K1215	Readiness and sustainment program	Train	002	
K2005*	Nuclear Propulsion Systems	Train	003	
K2010*	Principles of Bearings	Train	003	
K2015*	Principles of Clutches	Train	003	
K2020*	Principles of Shaft Lines	Train	003	
K2025*	Principles of Shaft Sealing and Supports	Train	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K2030*	Principles of Lubrication Systems	Train	003	
K2035*	Principles of Fuel Systems	Train	003	
K2040*	Principles of Cooling Systems	Train	003	
K2045*	Main Propulsion Design Considerations	Train	003	
K2050*	Operation of Main Propulsion Systems	Train	003	
K2055*	Principles of Steam Systems	Train	003	
K2060*	Operation of Auxiliary Systems	Train	003	
K2065*	Operation of Machinery Control and Surveillance Systems	Train	003	
K2070*	Principles of Degaussing/Deperming Systems	Train	002	
K2075*	Principles of Telephone Systems	Train	004	
K2080*	Operation of Electrical Systems	Train	003	
K2085*	Operation of HVAC Systems	Train	003	
K2090*	Effects of Damage on Operational Effectiveness and Ship's Safety	Train	002	
K2500*	Liquid Husbandry underway and alongside	Train	003	
K2505*	Docking policy and principles	Train	002	
K2510*	Docking Procedures	Train	002	
K2530*	Logistical Readiness	Train	002	
K2535*	Magnetic Signature	Train	002	
K2540*	Quiet Running States	Train	002	
K2550*	Quality Control/Quality Assurance	Train	002	
K2555*	Maintenance Action Forms (CF 1304)	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K2560*	Supply Demands (CF 2302)	Train	002	
K2620*	Burning and Welding Principles	Train	003	
K2655*	Personnel Policies and Procedures	Train	001	
K2675*	Military Personnel Policies	Train	001	
K2680*	Civilian Personnel Policies	Train	001	
K2685*	Maintenance Procedure Development and Management	Train	002	
K2690*	Configuration Management Organization	Train	002	
K2695*	Configuration Management Process	Train	002	
K1210	Fleet Support planning	Train	002	
K1215	Readiness and sustainment program	Train	002	
K2005*	Nuclear Propulsion Systems	Train	003	
K2010*	Principles of Bearings	Train	003	
K2015*	Principles of Clutches	Train	003	
K2020*	Principles of Shaft Lines	Train	003	
K2025*	Principles of Shaft Sealing and Supports	Train	003	
K2030*	Principles of Lubrication Systems	Train	003	
K2035*	Principles of Fuel Systems	Train	003	
K2040*	Principles of Cooling Systems	Train	003	
K2045*	Main Propulsion Design Considerations	Train	003	
K2050*	Operation of Main Propulsion Systems	Train	003	
K2055*	Principles of Steam Systems	Train	003	
K2060*	Operation of Auxiliary Systems	Train	003	
K2065*	Operation of Machinery Control and Surveillance Systems	Train	003	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K2070*	Principles of Degaussing/Deperming Systems	Train	002	
K2075*	Principles of Telephone Systems	Train	004	
K2080*	Operation of Electrical Systems	Train	003	
K2085*	Operation of HVAC Systems	Train	003	
K2090*	Effects of Damage on Operational Effectiveness and Ship's Safety	Train	002	
K2500*	Liquid Husbandry underway and alongside	Train	003	
K2505*	Docking policy and principles	Train	002	
K2510*	Docking Procedures	Train	002	
K2530*	Logistical Readiness	Train	002	
K2535*	Magnetic Signature	Train	002	
K2540*	Quiet Running States	Train	002	
K2550*	Quality Control/Quality Assurance	Train	002	
K2555*	Maintenance Action Forms (CF 1304)	Train	002	
K2560*	Supply Demands (CF 2302)	Train	002	
K2620*	Burning and Welding Principles	Train	003	
K2655*	Personnel Policies and Procedures	Train	001	
K2675*	Military Personnel Policies	Train	001	
K2680*	Civilian Personnel Policies	Train	001	
K2685*	Maintenance Procedure Development and Management	Train	002	
K2690*	Configuration Management Organization	Train	002	
K2695*	Configuration Management Process	Train	002	
K1210	Fleet Support planning	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K1215	Readiness and sustainment program	Train	002	
K2005*	Nuclear Propulsion Systems	Train	003	
K2010*	Principles of Bearings	Train	003	
K2015*	Principles of Clutches	Train	003	
K2020*	Principles of Shaft Lines	Train	003	
K2025*	Principles of Shaft Sealing and Supports	Train	003	
K2030*	Principles of Lubrication Systems	Train	003	
K2035*	Principles of Fuel Systems	Train	003	
K2040*	Principles of Cooling Systems	Train	003	
K2045*	Main Propulsion Design Considerations	Train	003	
K2050*	Operation of Main Propulsion Systems	Train	003	
K2055*	Principles of Steam Systems	Train	003	
K2060*	Operation of Auxiliary Systems	Train	003	
K2065*	Operation of Machinery Control and Surveillance Systems	Train	003	
K2070*	Principles of Degaussing/Deperming Systems	Train	002	
K2075*	Principles of Telephone Systems	Train	004	
K2080*	Operation of Electrical Systems	Train	003	
K2085*	Operation of HVAC Systems	Train	003	
K2090*	Effects of Damage on Operational Effectiveness and Ship's Safety	Train	002	
K2500*	Liquid Husbandry underway and alongside	Train	003	
K2505*	Docking policy and principles	Train	002	

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
K2510*	Docking Procedures	Train	002	
K2530*	Logistical Readiness	Train	002	
K2535*	Magnetic Signature	Train	002	
K2540*	Quiet Running States	Train	002	
K2550*	Quality Control/Quality Assurance	Train	002	
K2555*	Maintenance Action Forms (CF 1304)	Train	002	
K2560*	Supply Demands (CF 2302)	Train	002	
K2620*	Burning and Welding Principles	Train	003	
K2655*	Personnel Policies and Procedures	Train	001	
K2675*	Military Personnel Policies	Train	001	
K2680*	Civilian Personnel Policies	Train	001	
K2685*	Maintenance Procedure Development and Management	Train	002	
K2690*	Configuration Management Organization	Train	002	
K2695*	Configuration Management Process	Train	002	
* = MSE only				

ANNEX C – QS SCALAR

N/A

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

2. Intent of this course is to provide students the sea experience necessary to be successful at the next stage of NTO occupational training. The course includes 25 days aboard an HMC Ship with 15 days of sea time IOT foster appreciation of the operation and maintenance of HMC Ships.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**1.1**
Date: April 2014	**Who Initiated Change: OT DIV**
Requirement for the Amendment:	
Description of Changes: • "Board phase" added to course duration calculator. Pre-board (EC) and board (PC) require significant time and must be time accounted for and attributed to course conduct as they are formal assessments. • PO 001 – removed "describe defence ethics program" from standard statement as this is covered in CAFJOD. • PO 002 – removed "describe ship design process", "shipboard technical administration", "noise control requirements", "describe CMIS", "describe shipboard equipment health monitoring techniques" from standard statement as is not required knowledge of NEI level candidate. • PO 002 – removed "describe NaMMs" from standard statement as this is part of "ship engineering documentation". • PO 003 – removed "describe hull systems" from standard statement as info is covered in other POs • PO 003 – removed "describe integrated machinery control systems" from standard statement as RCN now uses IPMS. • PO 004 – removed "describe signature reduction" from standard statement and replaced with "describe signature management" in PO 002.	

- Combined EO 00.105 and EO 001.03 together (now EO 001.03) Most TPs taught extensively in NEC title of this EO ‘Career progression of NTOs & NCM Trades within the Engineering Departments

- EO 001.04 – time modified: 3 periods, 1 for lecture and 2 for fireside chat with CO

- EO 002.01 – time modified: 1 hr lecture, 4 hrs FMFCS Tour, 3 hrs Sub/CPF tour = 8hrs

- EO 002.02 – modified time from 2 periods to 1 period

- EO 002.03 – removed ‘Signature Mgmt’ info out of this EO and created its own EO 002.04. Also modified the time from 2 periods to 1 period. Title of this EO ‘Survivability and Damage Limiting Considerations’

- EO 002.04 – created new EO for Signature Management. Deemed to be a required of junior NTOs.

- EO 002.05 – This use to be NaMMS (added to EO 002.06) This EO is now ‘Ship Maintenance Org’ and has added to it an intro to DRMIS

- EO 002.06 – This EO use to be DRMIS and now ‘Ship Engineering Documentation’ an intro to NaMMs has been added to this EO

- EO 002.08 – new EO ‘Present Technical Info’

- EO 003.01 – Added TP4 (Liquid Cargo) from ‘Hull Systems’ to this EO ‘Hull Structures’

- EO 003.02 – removed ‘Hull Systems as most TPs already described in EO 002.03– this EO is now ‘Main Propulsion Systems’

- EO 003.03 – this EO use to be Main Propulsion & Ancillary Systems, they are now separate, this EO is just ‘Ancillary Systems’

- EO 003.04 – this has been modified to include both IMCS & IPMS examples. Now includes intro to health monitoring (which had its own EO 002.09)

- EO 003.06 – Auxiliary Systems – added TP 7 (Sanitary Systems) taken from ‘Hull Systems’

- Annex E – added OJPR info to clarify process and use of PRB.

- Annex G – added IAW MARCORD 9-30

- Annex K – modified to include 5 x ECs. Written Ashore exam now labelled PC 001 and board as PC 002. Instructors should make use of CFNES SO 424 (Progress review) to note progress and performance issues prior to initiating a TRB.

Change Tracking number:

Trg Mgr:

Version:	**1.2**
Date: July 2018	**Who Initiated Change: NTDC (P)**
Requirement for the Amendment:	
Description of Changes: The official NEI QSP and the one being used to instruct NEI are not aligned. NFS(A) (then CFNES) conducted an internal review of the QSP in 2015 which resulted in a QSP proposal to NPTG for approval in Feb 2016. The proposed QSP was returned to NTDC(P) for amendment on 17 Feb 2017, but the update and resubmit was overcome by other priorities until now. Meanwhile, both NFS proceeded to use the unapproved QSP understanding that was the latest/best version of the QSP and accepting the risk associated with using an unapproved document. The proposed QSP was amended to include original feedback as well as many points identified through the CAAR process by NFS(A)/(P). The following is summary of the major updates: • Sea Days: the QSP identifies the Afloat phase of the course to consist of at least 25 days aboard an HMC Ship, at least 15 of those days being Sea Days. • In chapter 1, the Afloat phase requirement has been augmented to clarify that if sea days cannot be provided to NEI candidates for exceptional circumstances (i.e: due to operational requirement) a waiver shall be obtain from NPTG. This was done to ensure HHQ visibility into availability of sea-going platforms for training. • Addition of EO 002.04 Describe Signature Management Requirements. This EO was removed during the 2016 QSPWB but no record of why it was removed which suggest it was missed. Review of the JBOS and indicated this topic should be covered to give both NTO trades adequate exposure. • PC 002 is better described including purpose of board and individual questions, conduct, timings and standard. The board guidance section explicitly describes the expectations of the candidates for every questions. • Addition to PO 004 of the OJT with EO 004.15 Describe the Fire Control System. Much like the EO 002.04, this topic was covered in the previous version of the QSP but was removed for unknown reasons. Discussing this with the NTO community lead to consensus that this topic is a minor (in terms of scope) but important addition compared to existing RADAR suite. • Addition of Annex K with the NEI board marking guide which demonstrate the marking expectations.	
Change Tracking number:	
Trg Mgr:	

Version:	**2.0**
Date: 1 July 2023	**Who Initiated Change:** HMCS Venture NTO Div
Requirement for the Amendment: Moving QSP to new format	
Description of Changes: QSP has been edited to move into the new format. No major change of content was made to chapter 1 and 2.	

Small edits made to organizational names, references were updated to reflect new references. No information was added to the QSP where there was not information in the previous QSP.
Change Tracking number:
Trg Mgr: Lt(N) Rodney Newcombe

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.	Classroom	Overhead projector, desktop or laptop computer.	1	1 additional classroom may be required for group activities. For the Oral Examination Board preparations, at a minimum, 2 classrooms should be reserved for concurrent practice boards. Board's schedule should be displayed on each door, so students are aware of timings. For this phase up to 3 classrooms will be required for 4 to 5 days, depending on the Oral Examination Board schedule and the number of candidates requiring boards.
NOTES:				

3. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.	N/A	N/A	N/A	N/A	N/A	N/A
	NOTES:					

4. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.	Ashore training handbook		1	
b.	Afloat Training Handbook		1	
NOTES:				

5. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.	Sea kit		1	
NOTES:				

6. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.	32 page exercise book (note book)		1	
b.	High-lighter		1	
c.	Red pen		1	
d.	Blue pen		1	
NOTES:				

7. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course:		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs		**$56.92**	**$0**	**$8531.76**	**$42 658.80**

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Assuming 50% of students are on TD and 50 % are posted to the area.

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT (Ashore Phase)	**Minutes**
Actual Training Time	2900
ATT Sub-total	**2900**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	400*
Training Administrative Time (TAT)	60
Training Travel Time (TTT)	50
Physical Fitness (PT)	N/A
Sub-total	**510**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**9 days**

*Extra time in BAT to account for initial sea kit issue

Afloat Phase	**Days**
Candidates will complete OJT aboard ship (to include 15 days at sea)	25

Allocation of ATT (Pre-Board Phase)	**Minutes**
Actual Training Time	1500
ATT Sub-total	**1500**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	NA
Training Administrative Time (TAT)	NA
Training Travel Time (TTT)	NA
Physical Fitness (PT)	NA
Sub-total	**0**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**4 days**

Allocation of ATT (Board Phase)	**Minutes**
Actual Training Time	1600
ATT Sub-total	**1600**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	NA
Training Administrative Time (TAT)	NA
Training Travel Time (TTT)	NA
Physical Fitness (PT)	NA
Sub-total	**0**

Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**5 days**

Course total	**Days**
Ashore phase	9
Afloat phase (including 15 days at sea)	25
Pre-Board phase	4
Board phase	5
Total	**43**

OPTIMUM TRAINING SCHEDULE

Week 1 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	Course Introduction (All Staff & CMDT)	**EO 001.03** Career Progression of Naval Eng.	**EO 002.01** Describe the Missions Assigned to Warships	**EO 002.01** Tour of FMFCS/CB	**EO 002.08** Present Technical Information
Period 2 **0850-0940**		**EO 001.03** "Bringing PROTECTEUR Home" documentary	**EO 002.01** Describe the Missions Assigned to Warships	**EO 002.01** Tour of FMFCS/CB	**EO 003.02** Main Propulsion Systems
Period 3 **0950-1040**	**EO 001.01** Development of Nav Eng. Occupations	**EO 001.03** Bringing PROTECTEUR Home" classroom chat	**EO 002.01** Describe the Missions Assigned to Warships	**EO 002.01** Tour of FMFCS/CB	**EO 003.02** Main Propulsion Systems
Period 4 **1040-1130**	Physical Training	**EO 001.04** Roles of Naval Engineering Officers	**EO 002.01** Describe the Missions Assigned to Warships	**EO 002.01** Tour of FMFCS/CB / Admin	Physical Training
Period 5 **1230-1320**	Administration	**EO 001.04** Fireside Chat with the Commandant	**Physical Training**	**EO 002.06** Shipboard Engineering Documentation	**EO 003.03** Ancillary Systems (Part 1)
Period 6 **1320-1410**	Administration	**EO 001.04** Fireside Chat with the Commandant	**EO 002.04** Signature Management Requirements	**EO 002.07** Docking and undocking Considerations	**EO 003.03** Ancillary Systems (Part 1)
Period 7 **1420-1510**	**EO 001.02** History of Naval Engineering	**EO 002.02** Factors Affecting Habitability	**EO 002.04** Signature Management Requirements	**EO 003.01** Hull Structures	**EO 003.04** Machinery Control Systems
Period 8 **1510-1600**	**EO 001.02** History of Naval Engineering	**EO 002.03** Survivability & Dmg Lmtg Considerations	**EO 002.05** Shipboard Maintenance Organization	**EO 003.01** Hull Structures	**EO 003.04** Machinery Control Systems

Week 2 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	**EO 003.05** Electrical Power Generation and Distribution	**EO 004.02** Command and Control Systems	**EO 004.05** Sonar Systems	**PC 001 (Ashore Exam)**	Sea Kit Issue/Dockyard Stores
Period 2 **0850-0940**	**EO 003.05** Electrical Power Generation and Distribution	**EO 004.02** Command and Control Systems	**EO 004.06** Navigation Systems		
Period 3 **0950-1040**	**EO 003.05** Electrical Power Generation and Distribution	**EO 004.03** Naval Weapons Systems	**EO 004.06** Navigation Systems	Course Critique	Sea Kit Issue/Dockyard Stores
Period 4 **1040-1130**	Administration	Physical Training	**EO 004.07** Communication Systems	Physical Training	Training Travel Time (return from Dockyard w/ Kit)
Period 5 **1230-1320**	**EO 003.06** Auxiliary Systems	**EO 004.03** Naval Weapons Systems	**EO 004.07** Communication Systems	Administration	Physical Training
Period 6 **1320-1410**	**EO 003.06** Auxiliary Systems	**EO 004.04** Radar Systems	**EO 004.08** Electronic Warfare Systems	Administration	Exam - rewrite
Period 7 **1420-1510**	**EO 004.01** Warfare System Requirements	**EO 004.04** Radar Systems	**EO 004.08** Electronic Warfare Systems	Course Admin/Sea Phase prep	Exam - rewrite
Period 8 **1510-1600**	**EO 004.01** Warfare System Requirements	**EO 004.05** Sonar Systems	**PC Review**		Exam - rewrite

Pre Board Time	Day 1	Day 2	Day 3	Day4	Day 5
Period 1 **0800-0850**	Pre Board Schedule/ Board Overview Brief	Drawing	Drawing	Drawing	MS Recap
Period 2 **0850-0940**	MLO/CMS	MLO/FCS	IPMS/CMS	Steering/W&S	
Period 3 **0950-1040**	MP&S/Bridge	MP&S/FCS	PG&D/W&S	MLO/Comms	MS Recap
Period 4 **1040-1130**	Physical Training	Admin	Physical Training	Admin	Physical Training
Period 5 **1230-1320**	MP&S/Comms	IPMS/FCS	MLO/Bridge	FCS	CS Recap
Period 6 **1320-1410**	MLO/W&S	FOS/Bridge	MP&S/Comms	MP&S/Comms	
Period 7 **1420-1510**	IPMS/FCS	IPMS/W&S	FOS	IPMS/Bridge	CS Recap

Board Week Time	Day 1	Day 2	Day 3	Day4	Day 5
Period 1 **0800-0850**	NEI Board	NEI Board	NEI Board	NEI Board	Return Sea Kit
Period 2 **0850-0940**					
Period 3 **0950-1040**	NEI Board	NEI Board	NEI Board	NEI Board	Return Sea Kit
Period 4 **1040-1130**					Admin
Period 5 **1230-1320**	NEI Board	NEI Board	NEI Board	NEI Board	Course Reports
Period 6 **1320-1410**					
Period 7 **1420-1510**	NEI Board	NEI Board	NEI Board	NEI Board	Course Graduation
Period 8 **1510-1600**					

AFLOAT PHASE TRACKING SPREADSHEET SAMPLES
WATCH ROTATION SAMPLE

1. The schedule below is based on 0800 hrs departure from home or foreign port.

	DD/MM/YY	DD/MM/YY	DD/MM/YY	DD/MM/YY	DD/MM/YY	DD/MM/YY
Middle 2330-0330		CCR/ Bridge Group 2	MCR Group 1	Ops Room Group 4	CCR/ Bridge Group 3	MCR Group 2
Morning 0330-0730		MCR Group 3	Ops Room Group 2	CCR/ Bridge Group 1	MCR Group 4	Ops Room Group 3
Forenoon 0730-1130	MCR Group 1	Ops Room Group 4	CCR/ Bridge Group 3	MCR Group 2	Ops Room Group 1	CCR/ Bridge Group 4
Afternoon 1130-1530	Ops Room Group 2	CCR/ Bridge Group 1	MCR Group 4	Ops Room Group 3	CCR/ Bridge Group 2	MCR Group 1
First Dog 1530-1730	CCR/ Bridge Group 3	MCR Group 2	Ops Room Group 1	CCR/ Bridge Group 4	MCR Group 3	Ops Room Group 2
Last Dog 1730-1930	MCR Group 4	Ops Room Group 3	CCR/ Bridge Group 2	MCR Group 1	Ops Room Group 4	CCR/ Bridge Group 3
First 1930-2330	Ops Room Group 1	CCR/ Bridge Group 4	MCR Group 3	Ops Room Group 2	CCR/ Bridge Group 1	MCR Group 4

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
PC 001	All POs	To assess candidate theoretical knowledge	Written assessment	At the culmination of the ashore phase	2 x 50 min	Nil		Pass/Fail 60%	Pass = Afloat phase 1st Fail = Re-test 2nd Fail = TRB NPTORD 4500-54
NEI Afloat Req Pack	OJPR Package onboard ship	To assess the performance outlined in the OJPR and application of the theoretical knowledge gained during Ashore Phase.	By CTO/EO aboard an HMC Ship using req pack.	Throughout OJPR	1) 25 training days (15 days at sea)	HMC Ships, CTOs (CS & MS), Afloat book.		Pass/Fail: Completion of OJPR Package IAW requirements standards; Annex E	Failure to maintain sufficient progress of OJPR as outlined in Annex E will result in progress review (NPTORD 4500-55 OJPR Management) at earliest indication. TRB may be convened based on outcome of progress review

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC 001	EOs 002.08, 002.09 002.10, 002.11, 002.12, 002.13, 003.07, 003.08, 003.09, 003.10, 003.11, 003.12, 003.13, 004.09, 004.10, 004.11, 004.12, 004.13, 004.14, 004.15,	To assess the candidate's ability to present technical information	Practical	During pre-board week.	1 x 50 min	Instructional staff to invigilate in a suitable testing environment		Pass/Fail IAW board marking guide and standard statement	First failure; rewrite Second failure;
PC 002	All POs	To evaluate the candidate's technical knowledge and ability to present information.	Oral Board IAW NAVORD 4500-20	On successful completion of pre-board	2 x 50 minute periods	Boardroom, Board Chair and Board members in accordance with applicable NAVORD		Pass/Fail IAW board marking guide – Annex L	Pass = Qualification granted 1st Fail = Re-assess 2nd Fail = TRB NPTORD 4500-54

ANNEX I – NEI BOARD ASSESSMENT CHECKLIST

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Boardroom.

Arrangements and Procedures

2. PC 002 will be a practical test based on content from all POs and is conducted IAW NAVORD 4500-20 and direction within this QSP.

3. Completion of OJPR is a prerequisite for PC 002.

4. Pass / Fail will be assessed by the board IAW this QSP.

5. Time required: 32 x 50 min period(s) to adequately allow time for all students to be tested as required.

6. References will not be required.

PART I – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from all POs.

2. During this practical assessment, you will be required to conduct an oral qualification board assessment. You must pass this assessment to be granted the NEI MITE code. To pass the test, you must complete the oral qualification board in accordance with the QSP and assessment rubric. References will not be provided.

3. Pass / Fail or Pass mark is assessed against the standard in this QSP

4. You will have 2 x 50 min period(s) to complete this assessment.

5. Do you have any questions before you start?

PART II – PERFORMANCE CHECK NOTES

1. The NEI Qualification Board is comprised of three questions. The first question is deemed to be critical and is weighted 4 out of 10 possible marks. The two remaining questions are weighted at 3 of 10 each.

2. Candidate shall achieve a minimum overall average mark of 60%, including a minimum of 60% for question 1 (critical question).

PART III – ASSESSMENT OF CANDIDATE

General

1. PC 002 is the culmination of the entire NEI course and all content learned or experienced during the Ashore and Afloat phases of the course are subject to questioning during PC 002. This board will be scheduled by the NEI Training Officer.

Standard of Performance

2. A hands-on approach to learning is a key element in establishing the broad base of experience required for this phase of training. The standard for this training has been revised over the years through observation and experience. To this end, the CTOs and board members are requested not to demand more of the candidate's presentations and pre-boards than is consistent with the standard established in this QSP. The items listed in the EOs describe the requisite level of detailed required.

Oral Examination Board Procedure

3. The Oral Examination Board conducted at the end of the NEI training will measure the candidate's understanding of the material studied as well as their public speaking skill (presentation). In order to pass the board, the candidate must receive a mark of at least 60%. The board shall be composed of a LCdr NTO (either MS Eng or NCS Eng HOD experienced), one Lt(N) MS Eng BOQ qualified, and one Lt(N) NCS Eng BOQ qualified.

4. During the board the candidate shall not have access to any references, and will be supplied with appropriate drawing material. The board shall consist of three (3) questions related to the content of NEI. The first question will involve a block diagram and presentation concerning one MS or one CS EO. The candidate will be informed which block diagram he/she is required to present 45 minutes prior to the board. During this time, they will produce any diagrams or speaking notes they wish to use in their presentation. The candidate will present to the members of the board, followed by a question period where they will be asked to expand on various aspects of the system or other familiarization tasks as appropriate with the standard of performance within this QSP. The remaining questions shall consist of a systems integration question or miscellaneous questions dealing with any topic or concept contained within the QSP.

5. In the event of a board failure or training difficulties, refer to chapter 3 and Annex K for further instructions.

Board Guidance

6. **Preparation**: initial preparation time 45 minutes for Q1 and Q2

7. **Marking**: the board should be scored with 80% of the mark for each question consisting of correct content and 20% on presentation. The pass mark for the board is 60%, constituting a combined average/weight of Q1, Q2 and Q3. This annex shall be used to record the board results.

8. **System Brief**: The initial brief given by the candidate for Q1 and Q2 should be no more than 15 minutes each. The level of detail expected to be presented is rudimentary and very basic. The brief should include a discussion on the following areas, as a minimum:

a. Purpose
b. Major Components
c. Modes of Operation
d. System Interfaces
e. Redundancy and Survivability

9. **General Expectations**: All answers to questions are expected to be:

- *Clear* (e.g. uses appropriate level of language, technical terminology, no colloquialisms/slang, etc.)
- *Concise* (e.g. answers are brief yet comprehensive)
- *Logical* (structures response sequentially)
- *Confident* (answered with conviction or no hesitation)
- *Accurate* (correctness of response appropriate to at least NEI level of knowledge)

10. **Board Timings**: the board should last less than 2 hours from when the board members introduce themselves to end.

<u>Q1. Major CSE Question (Weight 5)</u>

11. **Purpose**: to assess the candidate's rudimentary understanding of CSE systems as fitted onboard HMC Ship.

12. **Time**: 35 min, including questions

13. **Scope**:
a. Navigation
b. CMS
c. Weapons/Sensor Envelopes
d. RADAR Suite
e. Fire Control Suite
f. Internal and External Comms

14. **Expectations**: Correctly explain the purpose of the system as fitted in HMC Ships

- Correctly list the major system components
- Correctly describe the location of major system components (general area within compartment)
- Correctly describe the operation of major system components (basic)

- Correctly describe normal, abnormal and emergency operating modes (high level)
- Correctly provide significant capabilities & limitations of the system (ranges; air, surface or both, etc)
- Correctly explain survivability and redundancy features of the system design

Q2. Major MSE Question (Weight 5)

15. **Purpose**: to assess the candidate's rudimentary understanding of MSE system/suite as fitted onboard HMC Ship.

16. **Time**: 35 min, including questions

17. **Scope**:
    a. HVAC
    b. Seawater Services
    c. Main Propulsion and Shaft line
    d. Main Lube Oil System
    e. IPMS
    f. EPG&D
    g. Steering System
    h. Fuel Oil Services

18. **Expectations**: Correctly explain the purpose of the system as fitted in HMC Ships

- Correctly list the major system components
- Correctly describe the location of major system components (general area within compartment)
- Correctly describe the operation of major system components (basic)
- Correctly describe normal, abnormal and emergency operating modes (high level)
- Correctly provide significant capabilities & limitations of the system (i.e. for propulsion max & min ship's speed)
- Correctly explain survivability and redundancy features of the system design

Q3. Minor Question (Weight 3)

19. **Purpose**: to assess the candidate's individual experiences during the afloat phase of the NEI course.

20. **Time**: 20 min, including questions

21. **Scope**: Minor question should be selected based the limited sea time the candidate may have experienced. Examples may include:
    a. CSE watch makeup

b. MSE/MCR watch makeup
c. Combat Department watch makeup
d. Ops Room layout
e. Bridge layout
f. Pre-sail checks (purpose, some examples)
g. etc.

PART IV – LIST OF BOARDABLE SYSTEMS

MSE	**CSE**
Main Lubricating Oil (MLO)	Combat Management System (CMS)
Main Propulsion & Shaftline (MP&S)	Navigation Systems
Integrated Platform Management System (IPMS)	Communications Systems
Fuel Oil Services (FOS)	Fire Control System (FCS)
Power Generation & Distribution (OG & D)	Weapons & Sensors (W&S) Envelope
Steering	RADAR Suite

Candidate:	**School:**
Qualification: NEI	**Date:**
Chairperson:	**Board Member 1:**
Board Member 2:	**Course CTO**:

Instructions:
Each question is scored out of 10 for content and presentation which are weighted to create a score out of 10 for each question.[1] Each question is then weighted to create an overall board mark.[2] The candidate shall achieve a minimum overall board mark of 60%[3].

Score	Content	Score	Presentation
0	Unable to answer any part of the question or even begin to discuss the topic.	0	Many or more) major deficiencies
1	Unable to communicate a basic level of technical knowledge of engineering principles.	1	Some (0 to 2) major deficiencies
2	Able to answer minor parts of the main question but unable to answer follow up questions.	2	Many (3 or more) intermediate deficiencies
3	Displays a very low level of technical knowledge of engineering principles.	3	Some (0 to 2) intermediate deficiencies
4	Able to answer parts of the main question and a limited number of follow up questions displaying only a restricted level of technical knowledge.	4	Many (3 or more) minor deficiencies
5	Requires assistance answering the main question and requires prodding to answer follow up questions.	5	Some (0 to 2) minor deficiencies
6	Displays an acceptable level of technical knowledge of engineering principles and administration requirements.	6	Acceptable
7	Displays an above average level of technical knowledge of engineering principles.	7	Good
8	Displays a high level of technical knowledge of engineering principles.	8	Very Good

9	Generates discussion through the display of excellent technical knowledge of engineering principles.	9	Excellent
10	Displays an exceptional level of technical knowledge answering the main question and follow up questions well beyond the expected level.	10	Exceptional

Note 1	**Qualification**	**Content**	**Presentation**
	NEI	0.8	0.2
	BOQ	0.8	0.2
	HOD	0.6	0.4

Note 2	**Question**	**Weight**
	1	5
	2	5
	3	3

Note 3	Presentation is an aggregate evaluation of the following: - Dress and Deportment - Professionalism - Use of Proper Terminology - Interaction with Board Members - Eye Contact - Pace and Duration - Volume and Clarity of Voice - Body Language and Movement - Confidence and Conviction in Answers - Organization of the Presentation and Answers to Questions - Clarity of Board Drawings - Use of the Board/Drawing to Assist in Explanations.

Scoring Matrix:

Score (1 – 10)	Weight			
	0.2	**0.4**	**0.6**	**0.8**
1	2%	4%	6%	8%
2	4%	8%	12%	16%
3	6%	12%	18%	24%
4	8%	16%	24%	32%
5	10%	20%	30%	40%
6	12%	24%	36%	48%
7	14%	28%	42%	56%
8	16%	32%	48%	64%
9	18%	36%	54%	72%
10	20%	40%	60%	80%

QUALIFICATION:			CANDIDATE:		
DATE:			School:		
			CCA Rep Use Only		
QUESTION	CONTENT (0 – 10)	CONDUCT (0 – 10)	SCORE (S) (0 - 10)	WEIGHT FACTOR (WF) (3 - 5)	WEIGHTED SCORE (WS = WF x S)
1				5	
2				5	
3				3	
TOTALS:				13	

Mark (%): = TWS x 10 = TWF

Result: = Pass / Fail (Circle one)

Comments:

NOTE: Name and Signature are Mandatory

Name of Board Member **Signature**

PROTECTED B (when completed)

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	31 / 03 / 2023	OJPR Updated to new RCN QSP standards

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Training Officers.

OJPR SIGNATURE SHEET:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YY YY	OPI
PO 002 Describe the Operation and Maintenance of Shipboard Systems					
EO 002.08	Present Technical Information – System Pre-boards (Onboard & In class)	Conduct CS pre-board IAW EO 002.08 standard statement and technical information			CTO
		Conduct MS pre-board IAW EO 002.08 standard statement and technical information			CTO
		Extra Pre-boards (if time permits)			CTO
EO 002.09	Describe MSE Department Watches	MSEO daily rounds			MSEO
		MCR Watch – EOOW			CTO
		MCR Watch – Jnr EOOW			CTO
		MCR Watch – Jnr EOOW Operator			CTO
		MCR Watch – ETOW			CTO
		MCR Watch – Engineering Roundsperson			CTO
		MCR Watch – Engineering Roundsperson			CTO
		MCR Watch – BDCS			CTO
		MCR Watch – DC Roundsperson			CTO
		Presentation – MSE Department Overview			CTO
		Presentation – MSE Department At-Sea Organization			CTO
EO 002.10	Describe CSE Department Watches	CSEO daily rounds			CSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
		Ops Room Watch – Technician of the Watch			CTO
		Presentation – CSE Department Overview			CTO
		Presentation – CSE Department At-Sea Organization			CTO
EO 002.11	Describe Combat Department Watches	CCR Watch – NAVCOMM			CTO
		Ops Room Watch – TASOP			CTO
		Ops Room Watch – NESOP			CTO
		Ops Room Watch – NCIOP			CTO
		Ops Room Watch – ORO			CTO
		Bridge Watch – OOW/2OOW			CTO
		Presentation – Combat Department Overview			CTO
		Presentation – Combat Department At-Sea Organization			CTO
EO 002.12	Describe Shipboard Technical Readiness	Narrative – Technical Readiness			CTO
EO 002.13	Describe Naval Engineering Logistic Requirements	Supply Department Watch – Supply Technicians			CTO
PO 003 Describe the Operation of Marine Systems Engineering Equipment					
EO 003.07	Describe Hull Systems	Presentation – HVAC			CTO
		Presentation – Seawater Services			CTO
EO 003.08	Describe the Main Propulsion System & Shaftline	Presentation – Main Propulsion & Shaftline			CTO
EO 003.09	Describe the Main Lubricating Oil System	Presentation – MLO System			CTO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
EO 003.10	Describe the Integrated Platform Management System	Presentation – IPMS			CTO
EO 003.11	Describe the Power Generation and Distribution System	Presentation – PG&D Electrical Overview			CTO
EO 003.12	Describe the Steering System	Presentation – Steering System (Electro-Hydraulic)			CTO
EO 003.13	Describe the Fuel Oil Services System	Presentation – Fuel Oil Services System			CTO
PO 004 Describe the Operation of Combat Systems Engineering Equipment					
EO 004.09	Describe the Command and Control System	Presentation – CMS 330			CTO
EO 004.10	Describe the Weapon and Sensor Envelopes	Presentation – Weapon and Sensor Envelopes			CTO
EO 004.11	Describe the Navigation System	Presentation – Navigation System			CTO
EO 004.12	Describe the Radar Suite	Presentation – Radar Suite			CTO
EO 004.13	Describe the Internal and External Communication Systems	Presentation – Internal and External Communication Systems			CTO
EO 004.14	Describe the Electronic Warfare System	Presentation – Electronic Warfare System			CTO
EO 004.15	Describe the Fire Control System	Presentation – Fire Control System			

Progress Review Record

Authority & Period	Reviewed By	Signature	Date
Divisional Officer (Q1)			
Training Officer (Q1)			
Divisional Officer (Q2)			
Training Officer (Q2)			
Divisional Officer (Q3)			
Training Officer (Q3)			
Divisional Officer (Q4)			
Training Officer (Q4)			

TRAINING COMPLETION CERTIFICATE

I certify that_____has completed training to the standards stated in QSP – NEI, except as noted, and is recommended to challenge the Naval Engineering Indoctrination.

Date training completed:	
CTO	Date
XO	Date
CO	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Naval Engineering Indoctrination QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

EO	**COA 1** **TEL Delivery**	**COA 2** **Interim Delivery**	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

2. As the writing board for this course was run as only a QSPWB and not split into a QS and TP WBs, the info typically found here is located in Annex D.

RECORD OF AMENDMENT & AMPLIFICATION

3. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date: 31 / 03 / 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: Format of QSP into new format	
Description of Changes: • QSP updated to new RCN format • Boards were changed to new standard. Drawings were put back into the board, additional time given for the trainee to draw. • Organization names were edited to reflect new names	
Change Tracking number:	
Trg Mgr:	

ANNEX M – ACRONYMS AND ABBREVIATIONS

This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

Acronym and Abbreviation	Term or Definition
AA	Anti-Air
AAW	Anti-Air Warfare
AWW	Above Water Warfare
ACM	Acoustic Countermeasures
ADM (Mat)	Assistant Deputy Minister (Materiel)
ARPS	Acoustic Range Prediction System
ASuW	Anti-Surface Warfare
ASW	Anti-Submarine Warfare
BITE	Built-In Test Equipment
CBTO	Combat Officer
CCS	Command and Control System
CFAO	Canadian Forces Administrative Orders
CFTO	Canadian Forces Technical Order
CHOD	Chief of Department
CIWS	Close-In Weapons Systems
CRR	Combat Readiness Report
CSE	Combat Systems Engineering
CSEOTI	Combat System Engineering Officer Technical Instruction
CSEOTM	Combat System Engineering Officer Temporary
CWI	Continuous Wave Illuminator
DAOD	Defence Administrative Orders and Directives
DC	Damage Control
DRMIS	Defence Resource Management Information System
DWP	Docking Work Period
EC	Engineering Change
EDWP	Extended Docking Work Period
EHM	Equipment Health Monitoring
EOOW	Engineering Officer of the Watch
EOTI	Engineering Officer Technical Instruction
EW	Electronic Warfare
EWP	Extended Work Period
FTA	Formation Technical Authority
HAZMAT	Hazardous Material
HOD	Head of Department
HPR	High Priority Requirement
HVAC	Heating, Ventilation, and Air Conditioning
IFF	Identification Friend or Foe
IR	Infrared
LCMM	Life Cycle Material Manager
MARCORD	Maritime Command Order
MARE	Maritime Engineer
MARLANTORD	Maritime Forces Atlantic Order

MSE	Marine System Engineering
NaMMS	Naval Maintenance Management System
NEM	Naval Engineering Manual
NFP	Notice for Power
OCCAP	Oil and Coolant Condition Analysis Program
OPDEF	Operational Deficiency
OPSCHED	Operational Schedule (Changed to FLEETSCHED)
ORO	Operations Room Officer
PEL	Periodic Engineering Letter
PM	Preventative Maintenance
QHM	Queen’s Harbour Master
QR&O	Queen’s Regulations and Orders
RADHAZ	Non-Ionizing Radiation Hazard
SAR	Search and Rescue
SBT	Section Base Team
SOP	Standard Operation Procedure
SS	Ship’s Staff
SSE	Ship Systems Engineering
SSO	Ship’s Standing Orders
SWOAD	Ship Without Air Detachment
SWP	Short Work Period
SYO	Supply Officer
TEWA	Threat Evaluation Weapons Assignment
TMA	Target Motion Analysis
TRANREQ	Transfer Requirement
TRB	Training Review Board
UCR	Unsatisfactory Condition Report
VA	Vibration Analysis
WSB	Watch and Station Bill